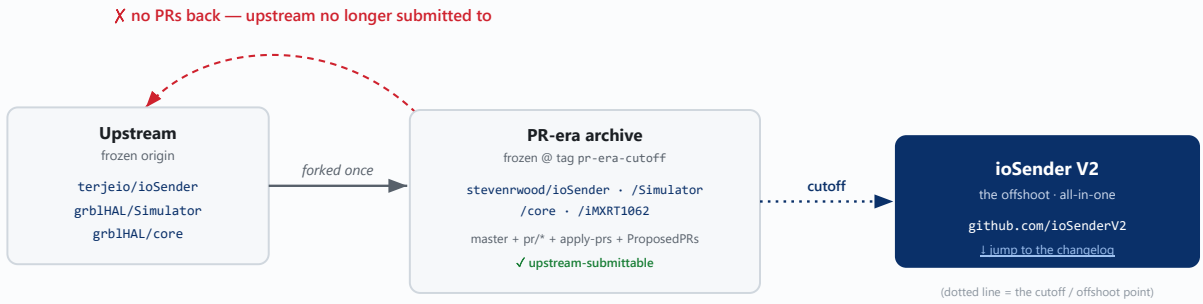
ioSender V2 — Overview

An enhanced **all-in-one** grblHAL / Grbl g-code sender. ioSender V2 is the ongoing offshoot of `stevenrwood/ioSender` (itself a one-time fork of `terjeio/ioSender`), severed from upstream at the **PR-era cutoff**. There are no composable feature branches and nothing is submitted upstream any more — the product is the `master` branch, consumed whole. This page maps how V2 came to be and how its repos wire together at runtime; the full changelog follows below.



One fork from each upstream → the PR-era archive (everything up to the cutoff, still submittable) → a dotted hand-off across the cutoff to **ioSender V2**, where all development now happens. Upstream is a frozen origin with the return path severed.

V2 repos & how they wire together at runtime

The app and simulator moved to the `ioSenderV2` org (original names kept, full git history preserved). The grblHAL **firmware** was *not* migrated — its deep submodule web makes it all-or-nothing, so it stays in the frozen fork, referenced by the `pr-era-cutoff` tag.

`ioSenderV2/ioSender`	The WPF sender app (this document lives here). Talks to the controller over USB serial or Telnet/WebSocket.
`ioSenderV2/Simulator`	**Option-matched simulator.** On connect the app reads the controller's build options → a 12-hex signature, then pulls a matching sim from this repo's `sim-<sig>` releases (a shared build cache), dispatching the `build-matched-sim` CI to produce one if absent. Every download is public; a token is only needed to *trigger* a build.
`stevenrwood/iMXRT1062` firmware · tag-only	The grblHAL firmware build (Teensy 4.1 driver superproject + `core` submodule at `srw/combined`, incl. the WCS-rotation transform fix) the app connects to. Stays in the fork — build branch `srw/local-build-config`, frozen reference at tag `pr-era-cutoff`.

THE FULL CHANGELOG — EVERY FEATURE & FIX IN THE V2 BUILD

Features & Fixes

Fork `stevenrwood/ioSender` · base branch `master` · **90 improvements in the `integration` build.** The fork ships as one all-in-one build — every change below is already in the `integration` branch and consumed wholesale. · all LF-normalized (`.gitattributes * text=auto eol=lf`).

Features at a glance — grouped by area. **NEW** new capability, **CHG** changes existing behaviour, **FIX** bug fix. Numbers link to the detail below.

Connectivity & reliability

Connect to USB / Ethernet / simulator, with a Connect/Reconnect menu, remembered host and opt-in prefer-network	NEW	15
Validate controller — streams a capability-tailored test in check mode (no motion), reports pass/fail + 3D preview	NEW	16
Restore main-window *position* across launches (was size only)	CHG	17
Survive connection loss / SD-card swap — guarded writes + auto-reconnect instead of crashing	FIX	4
Telnet / websocket modal-dialog UI deadlock	FIX	24
`-forgetNetwork` startup flag — ignore the saved connection target for one run and open the connect dialog (scan the network / pick any transport) instead of auto-connecting the remembered host; non-destructive (the saved value is untouched, and re-learned from the controller's `$I` on the next serial connect)	NEW	66
Demo-video capture aids (`-demomarker`) — an opt-in timestamped marker log (`SESSION_START` / `RUN_START` / `RUN_END` / `TIMELAPSE_ON/OFF`) plus a run-bar *Timelapse* toggle, so a video compositor can sync the app screen-recording to the machine-camera footage and time-compress the long cuts; hidden and near-free when off	NEW	67
OBS auto-record bridge & tool-change cut markers — with `-demomarker`, ioSender drives OBS Studio over obs-websocket (auto Start/Stop recording on program load / end, so a shoot needs no manual Record button); `RUN_START`/`RUN_END` now bracket the actual cutting, with a tool change emitting `RUN_END`…`RUN_START` around the M6. Plus `build.ps1` forwards trailing flags to the launched app	NEW	68
Per-Monitor V2 DPI awareness — the app reads the *live* per-monitor DPI (like Edge / VS Code) instead of a system-DPI value cached at logon, so it renders at the correct scale regardless of resolution / scale changes and is no longer thrown off by a stale DPI cache; adds `-debuglog=dpi` diagnostics	NEW	69

Jogging & main page

Configurable main page — Edit Main Page editor, pinnable flyouts, assignable jog panels, per-offset Go/Set buttons	NEW	9
Native Xbox / game-controller support — analog-stick jog, remappable buttons, live Controller tab	NEW	8
3D view: Ctrl+click to jog (retracts Z first) + per-axis $23 orientation fix	NEW	20
Keyboard jog keeps working when focus drifts off the Job view	CHG	22

High-DPI / 125–150% text-scaling layout	CHG	7
Assignable *Jog distance +/−* controller actions — step the jog distance preset (separate from jog speed)	NEW	35
Run-bar jog preset spinners — read-only *Dist* / *Feed* up/down selectors (wrap-around) on the Check line; startup splash stays on top so the full init sequence is visible	NEW	45
Drag a tab header left/right to reorder tabs — main bar and every sub-tab strip; the order persists across restarts (and agrees with the Edit Main Page editor)	NEW	60

Machine setup & settings

Machine Setup Wizard — guided first-run machine description (catalog, home-corner picker, per-axis table)	NEW	21
grbl:Settings — change-from-startup highlight, granular revert, per-Save restore points, bitfield tooltips	NEW	18
Stepper calibration (scratch) tab — long-span V-bit steps/mm	NEW	5
Safe-Z on Go To — lift / traverse / descend instead of a diagonal rapid	CHG	23
Editable G28/G30 predefined offsets (Set rapids to target, then stores)	CHG	26
Named G28 fixtures — a saved library of named machine positions in the G28 flyout (Set stores where you are, Go rapids back with Safe-Z); for recurring jigs & vices	NEW	70
Machine Setup restructure — Settings split (Grbl/App), new Tools tab, guided step-tabs + startup gate	NEW	28
Settings redesign — flat tab strip, free-text `$`-setting search (with match nav), grbl-settings autosave, dialog editors folded in as tabs	CHG	42
Per-user config folder + standalone `.xml` folded into `App.config` sections (`ConfigPanel<T>` wizards, curated defaults); console output search + font size, shared reset footer, bindable ResetAndUnlock	CHG	43
Lathe mode toggle — App › Main checkbox enables/disables lathe mode, adding or removing the Lathe wizards tab (restart to apply)	NEW	47
Restart-needed cue — the Restart button pulses red while a restart-only change is pending and offers to relaunch when you leave Settings	CHG	48
Connect: Network default + **Scan network** — finds grblHAL controllers on the LAN (confirms each by a `?`/`$I` handshake, not just an open port) and lists them in an editable host picker	NEW	59
grbl:Settings tree now populates when opened after connecting — an early activation during the connect handshake loaded the settings group-less (before `$I` reported enum support), so the grblHAL settings tree never built; it now fetches the group metadata once enums are known and rebuilds the tree on entry	FIX	61
… and the early activation itself is now suppressed at source — the Settings view ignores its inner tab-strip's startup selection churn until the view is actually shown, so `Activate` can no longer fire mid-handshake (belt-and-suspenders over #61)	FIX	65

Capability-gated tabs own their own “unavailable” reason — each gated view declares why it’s missing, so the removal and the *Edit Main Page* listing can’t drift (fixes the SD Card “needs SD” vs. “needs a filesystem” mismatch)	CHG	80
Restart prompt no longer stranded behind the relaunch splash when applying a restart-only change	FIX	81

Commissioning & workflow tools

Surface Spoilboard — machine-referenced facing (Z-only touch-off, Reuse Z0, finish pass, leveling check)	NEW	29
Load Stock — corner-probe stock size/flatness/skew/thickness, results popup, Copy size → Fusion add-in	NEW	30
Squareness (Auto Square) — drill an L, measure, write the ganged-axis squaring offset + re-home	NEW	31
Probe definition diagrams — per-type schematic in the probe editor labelling the measurements to enter	NEW	34
Probe input follows the selected probe type — tool setter → `Q1`, else the main probe → `Q0` (guards a stale selection)	FIX	36
Start Job (was Load Stock) — first tab, connection-gated tabs, auto 3D-probe at the front-left origin, writes `start_job.macro`; new Verify skew pass (re-touch each corner in the rotated frame); flyout persistence & offset-Set fixes	NEW	44
Tools tab guidance — each Tools tab (tool table, spoilboard surfacing, scratch calibration, Trinamic, PID) now shows an on-screen “About” explainer of what it does and how to use it	NEW	52
Start Job tab no longer strands + program view stays a right column — a probe/hold that didn’t cleanly clear could leave a gated tab disabled with no way back; the generated program overlay no longer grows to cover the whole tab	FIX	57
Probing & Tools sub-tabs share one look — instructions on the ambient grey, results in a white panel, consistent across the four Probing tabs and Stepper calibration	CHG	62
Check mode (`$C`) moved off the status bar to a checkable **Cycle Start** right-click item — a dry-run / validate toggle now sits with the button that starts a run	CHG	63
Smoother, faster corner probing — post-trigger back-offs are gentle feeds (no rapid-accel table shudder) and the Z probes assume ≤ 1″ stock for a much shorter seek	CHG	71
Start Job estimated stock size — setup fields relabelled *Est.* plus a new *Est. thickness (Z)* field that warns when stock exceeds 1″	NEW	72
Consolidate the Start Job corner probe to a single macro (pcorner) and add optional sacrificial spacer/backer support so a thin sheet is probed on the metal, not the backer.	NEW	85
With Measure on and the estimated size too large, corners that fall over bare spoilboard now abort at once instead of driving a doomed side probe into air.	FIX	86

Files, SD card & ATC

Load Folder — assemble per-toolpath `.nc` files into one outlined program (+ Fusion add-in)	NEW	6
SD + LittleFS browser & file manager — multi-filesystem, View/Edit/Create/Upload via Notepad	NEW	14
SD Card first-line cleanup — the filesystem status no longer overlaps the checkbox, and the free-space banner is hidden when the controller reports no sizes (was a bare “mounted”)	FIX	51
ATC tool-change macro provisioning — Install ATC macros + seeded Start Job	NEW	25
On an ATC controller the setup gate could dump you at step 6 even though the macros were installed and turned green a moment later	FIX	88

Program view & G-code

Reusable program view — file / folder / each wizard gets its own view; one Cycle Start runs the active one	CHG	32
Background file / folder loading — big programs parse on a worker thread; the UI stays responsive, rows fill as read	CHG	32
Continuous block numbering — explicit `N` words shown in the Block column and hidden from the G-code column	CHG	32
Explain G-code on hover — plain-language per line/selection (decodes parameters, O-word CALLs, expressions); click the balloon to copy	NEW	33
One in-place streaming path — every wizard/macro runs in its own view; a run never overwrites the loaded job	CHG	37
Live 3D carve view — watch the stock get machined away (program-sized stock, real cutter profile flat/ball/V-bit, Play / Cycle Start playback) instead of just toolpath lines	NEW	41
3-line run view — on Cycle Start a generated program collapses to previous / current / next line at the bottom of the tab; click the title bar to expand	NEW	76

Macros, console & keys

Macro manager dialog (grid, F-key assign, confirm-on-run flag) + run-only macro flyout	NEW	10
Macro pre-processor directives — PREREQ / MBOX / WAITIDLE / PROMPT / `@path`	NEW	11
In-app Key Mappings editor + console state persistence	NEW	3
Console: inline terminal command prompt + output right-click menu (Clear / Save)	NEW	19
Bind any main + second-level tab to a key — right-click Bind-to-Key + header badges	NEW	79

Polish, onboarding & docs

Numeric fields select their value on focus — tab or click into any number field and just type the replacement (no clearing first); single-digit edits still work	CHG	58
High-DPI polish — fixed pixel sizes converted to Min/Auto so text doesn’t clip at 125–150% scaling	FIX	46

Lathe wizards rebuilt to the app layout idiom — the 5 upstream absolute-Margin wizards re-hosted in scaling StackPanel/Grid so they no longer clip/overlap at 125–150%	CHG	49
Tab strips fill their width — a StretchTabControl spreads each tab header proportionally across the row so tabs read as distinct (main bar + Settings / Tools / Lathe / Probing / Machine Setup)	CHG	50
Online user manual + in-app context help — `F1` (and the Help menu) open a task-oriented manual with novice / intermediate / machinist reading tracks, A–Z index, search, diagrams and screenshots; hosted at `iosenderv2.github.io/ioSender`	NEW	56
Onboarding tooltips — plain-language controller-state field + per-setting help	NEW	12
UI localization — all UI strings registered for 7 locales (cross-cutting)	NEW	9,10,12,14–16,18,19,21,23
Center owner-less modal dialogs (MPGPending, ProbeVerify)	FIX	27
Show Motor Fault / Warning signals in the Signals display	FIX	2
Build documentation — Mega-XL upgrade notes + repeatable tool-change guide	NEW	13
Assorted released-ioSender bug fixes (RP.Math build, LF endings, CSV null-ref, boolean settings, null-comms)	FIX	1
Headless build — `build.ps1` + VS Code tasks build/launch without opening the VS GUI (MSBuild found via vswhere); unhandled exceptions now dump to `%AppData%\ioSender\ioSender.crash.log` and exit `0xFA11` instead of blocking on a dialog	NEW	53
Opt-in diagnostic trace log — launch with `-debuglog` (or `IOSENDER_DEBUGLOG`) to write a timestamped flow trace to `%AppData%\ioSender\ioSender.debug.log`; off by default, so `DebugLog.Write(…)` call sites can live permanently in the code without a rebuild toggle	NEW	64
Code renamed LoadStock → Start Job to match the UI throughout (files, classes, `ViewType`, layout keys, config section)	CHG	54
Localization catch-up — later V2 features (Start Job, probing-redesign dialogs, main-page editor, Tools guidance, …) were `x:Uid`'d but had no LocBaml rows; every missing string backfilled into all 7 locale CSVs	NEW	55
Playbooks — the repeatable dev procedures (changelog entry, PDF/manual publish, builds, localization, firmware, demo video) collected into `docs/playbooks/`, one per file with ready-to-run steps	NEW	73
Startup dialogs no longer hidden behind the splash — the progress splash docks to the top so the connect dialog, message boxes and setup prompts stay visible	FIX	74
Status-bar messages shown at double height for 10 s when written, then shrink back — so an important message in the easily-missed status line gets noticed	NEW	75

Selected tab label goes bold and slightly larger — the active tab reads at a glance, not just by its white background (every StretchTabControl bar)	CHG	77
Run-bar jog pad restored — it had been squeezed to nothing by the Program button; run bar rebalanced into four columns with the button row filling the slack up to the jog pad	FIX	78
The deterministic dev playbooks are now one-command scripts — push to both remotes, regenerate the Overview PDF, run **gh**, and generate a changelog entry — so they are invoked, not re-derived each time	NEW	82
One live instance, a clean supervised restart, and opt-in crash reports with a minidump.	NEW	83
File menu dissolved onto the program view + right-click menu; toolbar row removed; Camera menu made opt-in.	CHG	84
Optional localhost server that drives the WPF UI by x:Uid so a change can be verified end to end without a human clicking	NEW	87
The optional test server grew from a proof slice into a real automation harness, and moved into its own app-agnostic assembly	NEW	89
The app-agnostic UI test server moved out to its own public repo and NuGet package; ioSender now consumes it as a package.	CHG	90

1	Bug Fixes for released ioSender	Files: 5 0 0 Lines: +28 -10
2	Surface Motor Fault / Warning signals	Files: 2 0 0 Lines: +5 -32
3	Console persistence + Key Mappings editor	Files: 12 0 +4 Lines: +1224 -15
4	Connection-loss handling + auto-reconnect	Files: 10 0 0 Lines: +522 -63
5	Stepper calibration (scratch) tab	Files: 13 0 +3 Lines: +773 -2
6	Load Folder + Fusion batch-post add-in	Files: 28 0 +8 Lines: +1634 -8
7	High-DPI / large-text layout	Files: 27 0 0 Lines: +126 -109
8	Xbox controller + Key Mappings editor polish	Files: 14 0 +4 Lines: +2255 -687
9	Configurable main page + flyouts + jog panels (XL)	Files: 31 0 +20 Lines: +2835 -119
10	Macro manager + run-only flyout	Files: 16 0 +2 Lines: +598 -26
11	Macro Processor Enhancements	Files: 8 0 +1 Lines: +723 -21

12	Onboarding tooltips	Files: 14 0 0 Lines: +283 -28
13	Build documentation (Mega-XL notes + tool-change guide)	Files: 1 0 +8 Lines: +2662 0
14	SD + LittleFS file browser & file manager	Files: 12 0 +3 Lines: +945 -135
15	USB / Ethernet / simulator connection support + Connect menu	Files: 26 0 +3 Lines: +1151 -68
16	Validate controller (check-mode g-code conformance test)	Files: 19 0 +1 Lines: +1208 -38
17	Restore main window position across launches	Files: 2 0 0 Lines: +60 -7
18	grbl:Settings — live edits, change highlight, revert & snapshots	Files: 12 0 +2 Lines: +659 -43
19	Console — inline terminal-style command prompt	Files: 13 0 0 Lines: +220 -21
20	3D view — Ctrl+click to jog + per-axis orientation	Files: 5 0 0 Lines: +206 -60
21	grbl:Settings — Machine Setup Wizard	Files: 12 0 +3 Lines: +1740 -1
22	Keyboard jog — keep active when Job-view focus drifts	Files: 2 0 0 Lines: +23 -1
23	Go To — optional safe-Z (lift, traverse, descend)	Files: 10 0 0 Lines: +54 -8
24	comms — fix telnet/websocket UI deadlock	Files: 2 0 0 Lines: +29 -10
25	ATC tool-change macro provisioning (Install ATC + Start Job)	Files: 7 0 +5 Lines: +804 -10
26	Offsets — editable G28/G30 (Set rapids & stores)	Files: 1 0 0 Lines: +21 -2
27	Dialogs — center owner-less modal dialogs	Files: 4 0 0 Lines: +4 -2
28	Machine Setup restructure + Tools tab + setup gate	Files: 4 0 +2 Lines: +237 -21
29	Surface Spoilboard — machine-referenced facing tool	Files: 2 0 +2 Lines: +673 0
30	Load Stock — corner-probe stock measurement	Files: 7 0 +7 Lines: +1125 0
31	Squareness (Auto Square) — ganged-gantry offset tool	Files: 2 0 +2 Lines: +686 0
32	Program view as a reusable object + background loading	Files: 20 0 +4 Lines: +830 -184

33	Explain G-code on hover (novice tooltips)	Files: 4 0 +1 Lines: +805 -39
34	Probe definition diagrams	Files: 2 0 0 Lines: +142 -22
35	Jog distance +/− controller actions	Files: 4 0 0 Lines: +17 -2
36	Probe input follows the selected probe type	Files: 3 0 0 Lines: +16 0
37	One in-place streaming path (stayPut removed)	Files: 3 0 0 Lines: +62 -110
38	Match the bundled simulator to the connected controller	Files: 2 0 0 Lines: +355 0
39	Re-establish modal state on a mid-program start	Files: 1 0 0 Lines: +22 0
40	Load Stock “apply work rotation” defaults off	Files: 1 0 0 Lines: +6 -1
41	Live 3D toolpath + material-removal carve view	Files: 4 0 +2 Lines: +1107 -62
42	Settings redesign — flat tab strip + searchable Grbl settings	Files: 23 2 0 Lines: +935 -542
43	Per-user config folder + standalone config folded into App.config	Files: 27 0 +3 Lines: +1269 -369
44	Start Job workflow + Verify skew, flyout & offset fixes	Files: 17 0 0 Lines: +369 -182
45	Run-bar jog preset spinners + startup splash + MDI Send align	Files: 4 0 +2 Lines: +132 -9
46	High-DPI audit — text no longer clips at 125–150% scaling	Files: 63 0 0 Lines: +196 -197
47	Lathe mode enable/disable checkbox	Files: 3 0 0 Lines: +56 -0
48	Restart button pulse + restart-on-leave prompt	Files: 2 0 0 Lines: +58 -1
49	Lathe wizards rebuilt to the app layout idiom (high-DPI)	Files: 5 0 0 Lines: +461 -387
50	Tab strips fill their width (StretchTabControl)	Files: 7 0 +1 Lines: +137 -14
51	SD Card first-line cleanup (overlap + free-space banner)	Files: 2 0 0 Lines: +9 -4
52	Per-tool guidance text on the Tools tabs	Files: 7 0 0 Lines: +46 -14
53	Headless build script + crash logging	Files: 2 0 +2 Lines: +255 -8

54	Rename LoadStock → Start Job (code matches the UI)	Files: 30 0 0 Lines: +177 -177
55	Localization sweep — V2 strings into all 7 locales	Files: 29 0 0 Lines: +1581 -1
56	Online user manual + in-app context help (F1)	Files: 2 0 +1 Lines: +126 -2
57	Fixes: Start Job tab strand + program-view overlay width	Files: 3 0 0 Lines: +26 -1
58	Numeric fields select their value on focus	Files: 1 0 0 Lines: +35 -0
59	Connect: Network default + LAN controller discovery	Files: 9 0 +1 Lines: +527 -9
60	Drag to reorder tabs (persisted across restarts)	Files: 6 0 +1 Lines: +306 -8
61	grbl:Settings tree populates after connecting	Files: 2 0 +0 Lines: +37 -4
62	Probing/Tools sub-tab layout standardized	Files: 6 0 +0 Lines: +38 -30
63	Check ($C) moved to Cycle Start context menu	Files: 11 0 +0 Lines: +48 -29
64	Opt-in diagnostic trace log (-debuglog)	Files: 3 0 +1 Lines: +179 -5
65	grbl:Settings premature activation prevented at source	Files: 1 0 +0 Lines: +19 -0
66	Startup flag -forgetNetwork	Files: 1 0 +0 Lines: +13 -0
67	Demo-video capture markers (-demomarker)	Files: 5 0 +1 Lines: +219 -1
68	OBS auto-record bridge + tool-change cut markers	Files: 5 0 +1 Lines: +211 -3
69	Per-Monitor V2 DPI awareness	Files: 3 0 +1 Lines: +79 -0
70	Named G28 fixtures	Files: 12 0 +1 Lines: +336 -3
71	Smoother, faster corner probing	Files: 2 0 +0 Lines: +30 -22
72	Start Job: estimated stock size & Z check	Files: 10 0 +0 Lines: +47 -16
73	Playbooks: one-stop procedure library	Files: 0 0 +14 Lines: +456 -0
74	Startup dialogs no longer hidden behind the splash	Files: 3 0 +0 Lines: +18 -17

75	Status-bar messages shown at double height	Files: 1 0 +0 Lines: +37 -0
76	Program view: 3-line run view	Files: 4 0 +0 Lines: +170 -2
77	Selected tab: bold, larger label	Files: 1 0 +0 Lines: +62 -1
78	Run-bar jog pad restored & row rebalanced	Files: 3 0 +0 Lines: +42 -39
79	Bind any tab to a key — right-click Bind-to-Key + header badges	Files: 13 0 +1 Lines: +716 -17
80	Capability-gated tabs own their own “unavailable” reason	Files: 12 0 +1 Lines: +129 -93
81	No more restart prompt hiding behind the relaunch splash	Files: 1 0 +0 Lines: +18 -1
82	Dev playbooks distilled to one-command scripts	Files: 5 0 +4 Lines: +384 -11
83	Relaunch supervisor, single-instance & crash reporting	Files: 5 0 +2 Lines: +850 -58
84	Main-menu overhaul	Files: 35 0 +0 Lines: +458 -65
85	Start Job: one probe macro + spacer support	Files: 19 2 +0 Lines: +80 -232
86	Start Job: fail fast over bare board	Files: 1 0 +0 Lines: +12 -4
87	UI test server (-testserver)	Files: 2 0 +1 Lines: +505 -1
88	Fix false Machine Setup step 6 (ATC macros)	Files: 2 0 +0 Lines: +17 -4
89	UI test server â€” full harness + independent assembly	Files: 96 0 +7 Lines: +1832 -241
90	UI test server — standalone repo + NuGet package	Files: 4 5 +0 Lines: +5 -1391
	Totals (90 changes)	Files: 877 9 +133 Lines: +40623 -6182

Cross-repo: which ioSender features the `stevenrwood/Simulator` fork exercises.

No build dependency, and all-or-nothing — you run the whole `Simulator@integration` build against ioSender (not a chosen subset), so this table is informational rather than something to satisfy. The coupling is runtime / protocol: the simulator only exercises a feature when the ioSender change below drives it over the socket.

ioSender feature	Simulator feature it lights up	Coupling

#6 Load Folder	3D carve view	`PushHeaderToSimulator()` sends the program's leading `(STOCK …)` / `(TOOL …)` comments so the carve knows stock size + cutter geometry.
#11 macro pre-processor	littlefs macros	`O<name> CALL` forwarding resolves macros stored on the simulator's `/littlefs`.
#14 SD / filesystem browser	littlefs, YModem	File browser + YModem uploader that puts files onto the simulator filesystem.
#15 connect-to-simulator	3D carve view, settings persistence, sim core	Connects to the standalone sim; *always-send-comments* forces `(TOOL)`/`(STOCK)` through for the carve; "My Machine" EEPROM uses the sim's settings persistence.
#25 ATC macro provisioning	YModem, littlefs, probe surfaces	"Install ATC macros" uploads via YModem to `/littlefs`; the Start-Job / `probe_tfl` cycle drives the modelled probe surfaces.

For the full virtual-CNC experience, run **`ioSender@integration`** against **`Simulator@integration`** — one all-in-one bundle each (the firmware-submodule fixes ride along inside the sim). The matched-simulator feature (#38) builds an option-matched sim automatically.

#1 Bug Fixes for released ioSender

File	+	−
`.gitattributes`	3	2
`CNC Controls/CNC Controls/JobControl.xaml.cs`	7	0
`CNC Controls/CNC Controls/Widget.cs`	3	1
`CNC Core/CNC Core/CNC Core.csproj`	6	0
`CNC Core/CNC Core/Grbl.cs`	9	7

A bucket of fixes for bugs that exist in released ioSender, independent of the feature PRs. Each is small and self-contained:

- **RP.Math build fix.** CNC Core compiles `GCodeEmulator.cs`, which uses the `RP.Math.Vector3` type, but the project file declared no reference to RP.Math, so a clean build fails to resolve the type. Adds the `RP.Math` and `RP.Math.Vector3` references using the *same* `HintPath` convention already used by `CNC Controls.csproj` (`..\..\Vector-master\RP.Math.Vector3\bin\Release\`). Purely additive.
- **LF line-ending guard.** Adds `.gitattributes` (`* text=auto eol=lf`) so line endings are normalized.
- **CSV null-reference fix.** Avoids a `NullReferenceException` when an alarm/error/settings-group CSV resource is missing.
- **Boolean settings convention.** The settings editor checkbox initialised with `value == "1"`, but a boolean is true when `value ≠ "0"` everywhere else (the list, controller-value parsing). Booleans reported as a truthy value other than the literal `"1"` showed enabled in the list yet left the checkbox unchecked. Canonicalises boolean values to `"1"`/`"0"` in `GrblSettingDetails.Value` and reads the checkbox with the `≠ "0"` convention.
- **Null-comms streaming guard.** `JobControl.ResponseReceived` is raised by a specific comms instance but writes to the static `Comms.com`; during a reconnect/teardown (startup handshake or relaunch) that static can be null while an in-flight response from the old link still arrives, NREing in the SendMDI/Reset cases. Bails out early when `Comms.com` is null.

#2 Surface Motor Fault and Motor Warning signals in Signals display

File	+	−
`CNC Controls/CNC Controls/SignalsControl.xaml`	2	30

Removes the capability-gating `Styles` that prevented the Motor Fault (**F**) and Motor Warning (**W**) indicators from ever showing.

grblHAL does not advertise these as `NEWOPT` capability tokens, so the visibility gates were effectively dead code — the LEDs were permanently hidden. Motor Fault / Warning are now always displayed alongside the other signal LEDs (H / S / P), turning red when the corresponding signal triggers. This surfaces the Motor Fault signal as intended, without requiring the firmware to advertise a capability flag.

#3 Persist console state and add an in-app Key Mappings editor

File	+	−
CNC Controls/CNC Controls/AppConfig.cs	14	0
CNC Controls/CNC Controls/AppConfigView.xaml	1	1
CNC Controls/CNC Controls/AppConfigView.xaml.cs	16	3
CNC Controls/CNC Controls/CNC Controls.csproj	7	0
CNC Controls/CNC Controls/ConsoleControl.xaml	3	3
CNC Controls/CNC Controls/ConsoleControl.xaml.cs	14	0
CNC Controls/CNC Controls/ConsoleWindow.xaml.cs	47	0
CNC Controls/CNC Controls/KeyMapEditor.xaml	115	0
CNC Controls/CNC Controls/KeyMapEditor.xaml.cs	606	0
CNC Core/CNC Core/CNC Core.csproj	7	0
CNC Core/CNC Core/KeypressHandler.cs	111	0
CNC Core/CNC Core/ShortcutKey.cs	136	0
ioSender XL/ioSender XL/App.config	6	0
ioSender XL/ioSender XL/MainWindow.xaml.cs	69	3
ioSender/ioSender/MainWindow.xaml.cs	69	3

Persists the console's state across sessions and replaces the old App Settings "*Save key mappings*" button (which only dumped the current bindings to `KeyMap<mode>.xml` for hand-editing) with a proper in-app **Edit Key Mappings** dialog. Applies to both **ioSender** and **ioSender XL**.

Console persistence: the three console checkboxes (*Verbose*, *Filter out realtime responses*, *Show all realtime responses*), whether the floating console window is open, and that window's size/position (clamped to the visible desktop so it can never restore off-screen) are all remembered. A new `ConsoleShortcut` setting (default `Esc`) toggles the console; it is now a first-class action inside the mappings editor rather than a separate, ad-hoc tab. A shared `ShortcutKey` helper (parse / storage / display) is used by both main windows, and an `AppConfig.ConsoleShortcutChanged` event re-registers the key live without a restart.

Key Mappings editor (`KeyMapEditor`, modal so jog/realtime dispatch can't fire mid-capture):

- One grouped, collapsible outline of every bindable action (grouped by function, like the Load Folder outline; groups remember their expansion per session). Click a row, press a combo to assign; right-click for *Reset to default* / *Clear*.
- Bindings changed from the factory default are highlighted; group headers flag a modified group and offer *Expand all / Collapse all / Reset group*.
- Per-axis jog keys are folded into the same list; rows and group headers carry tooltips.
- A *No-numpad keys* preset remaps the NumPad jog/step/feed actions to `Ctrl+1–8` and `[ ] − =` for keyboards without a numeric keypad (Windows exposes no reliable API to detect numpad presence, so this is a user-triggered preset).

The `KeypressHandler` gains an editor API (`GetActionBindings/GetJogBindings/Apply…`) over its existing function catalog, identifying bindings by reflection method name with a friendly-label map. Persistence still uses the existing `SaveMappings/LoadMappings`.

#4 Fix unhandled exceptions on connection loss (e.g. SD card swap)

File	+	−
CNC Controls/CNC Controls/AppConfig.cs	5	0
CNC Controls/CNC Controls/SDCardView.xaml.cs	13	0
CNC Core/CNC Core/Comms.cs	74	0
CNC Core/CNC Core/EltimaStream.cs	6	0
CNC Core/CNC Core/Grbl.cs	13	4
CNC Core/CNC Core/GrblViewModel.cs	45	0
CNC Core/CNC Core/SerialStream.cs	179	42
CNC Core/CNC Core/TelnetStream.cs	105	7
CNC Core/CNC Core/WebsocketStream.cs	79	8

When the controller's serial/USB device disappears — for example a board that resets and re-enumerates when its SD card is removed or reinserted — the status-poll timer kept writing to the dead port. The write threw (an `IOException`, then an `InvalidOperationException`: "BaseStream only available when the port is open") on a `System.Timers.Timer` thread, and the unhandled exception terminated the whole process. Opening the SD card tab afterwards threw similarly from `PurgeQueue`.

Changes:

- Guard all stream writes (`IsOpen` + try/catch). A device-gone fault now closes the port and starts an auto-reconnect instead of crashing.
- Add a shared, transport-agnostic `Reconnector` (used by serial, network and websocket) exposing `ConnectionLost` / `Reconnected` events. `GrblViewModel` handles them (stop/resume polling, show status); it is wired up at connect time in `AppConfig`.
- **Serial:** refactor open into a reusable `OpenPort()` that probes `GetPortNames()` and re-opens; publish the port only once it is actually open, to avoid a non-null-but-closed race; make `PurgeQueue` tolerant of a closed port; harden the receive handler against a port torn down mid-read.
- **Telnet / websocket:** guarded writes, loss detection (read returning 0 / `OnClose`) and reconnect via the same helper.
- Wrap the poll-timer callback in try/catch as a last line of defence.
- Guard the SD card view against acting on a closed connection.

#5 Add Stepper calibration (scratch) tab to grbl:Settings

File	+	−
CNC Controls/CNC Controls/CNC Controls.csproj	7	0
CNC Controls/CNC Controls/GrblConfigView.xaml	3	0
CNC Controls/CNC Controls/IGrblConfigTab.cs	1	0
CNC Controls/CNC Controls/StepperCalibrationScratchWizard.xaml	92	0
CNC Controls/CNC Controls/StepperCalibrationScratchWizard.xaml.cs	486	0
.gitattributes	3	2
CNC Controls/CNC Controls/LibStrings.xaml	1	0
CNC Core/CNC Core/CNC Core.csproj	6	0
ioSender XL/ioSender XL/App.config	6	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	168	0

Adds a new tab to grbl:Settings for V-bit *scratch-line* steps/mm calibration over a long span — a more accurate alternative to the short jog-based calibration for the X and Y axes.

It generates an `.nc` program that scratches paired lines (perpendicular to the calibrated axis) for several candidate steps/mm values. Because GRBL cannot change `$100`/`$101` mid-program, each candidate is simulated using the *commanded-distance trick*: the commanded distance `C = span × candidate / current`, so the machine physically travels the distance that the candidate steps/mm value *would* produce.

The program is loaded into the job view (with an optional save to disk). After cutting, the user measures the actual spacings; the tab then computes and saves the corrected `$100`/`$101` from those measurements (an implied value per pair, averaged). The prolog mirrors the Load Folder convention. X/Y only — Z continues to use the existing jog wizard.

Includes a small follow-up commit with high-DPI layout tweaks for the new tab.

#6 Add "Load Folder": load a folder of per-toolpath .nc files as one outlined program

File	+	−
.gitattributes	3	2
CNC Controls/CNC Controls/CNC Controls.csproj	2	0
CNC Controls/CNC Controls/FolderPicker.cs	159	0
CNC Controls/CNC Controls/FusionFolderLoader.cs	353	0
CNC Controls/CNC Controls/GCode.cs	156	0
CNC Controls/CNC Controls/GCodeListControl.xaml	45	0
CNC Controls/CNC Controls/GCodeListControl.xaml.cs	129	0
CNC Controls/CNC Controls/JobControl.xaml.cs	4	0
CNC Core/CNC Core/GCodeJob.cs	41	5
CNC Core/CNC Core/GrblViewModel.cs	7	1
Fusion Addin/README.md	116	0
Fusion Addin/install-macos.sh	39	0
Fusion Addin/install-windows.ps1	40	0
Fusion Addin/ioSenderBatchPost/ioSenderBatchPost.manifest	11	0
Fusion Addin/ioSenderBatchPost/ioSenderBatchPost.py	476	0
ioSender XL/ioSender XL/MainWindow.xaml	1	0
ioSender XL/ioSender XL/MainWindow.xaml.cs	5	0
ioSender/ioSender/MainWindow.xaml	1	0
ioSender/ioSender/MainWindow.xaml.cs	5	0
CNC Controls/CNC Controls/LibStrings.xaml	1	0
CNC Core/CNC Core/CNC Core.csproj	6	0
ioSender XL/ioSender XL/App.config	6	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	21	0
Locale/*/csv/ioSender.resources.*.csv (7 locales)	7	0

File > Load Folder reads a folder of per-operation files named `<seq>_<name>_T<tool>.nc` (as produced by the companion Fusion 360 add-in) and assembles them in memory into a single program:

- strips each file's header and program-end footer;
- inserts a `G53 G0 Z0` safe-Z retract + `M6 T<n>` tool change before each toolpath, but only when the file does not already contain an `M6` (so any post works);
- optionally restores rapid moves that Fusion Personal Use downgraded to `G1`;
- shows the result as a collapsible per-toolpath outline (groups default collapsed, and stay expanded while scrolling);
- adds *Start from this toolpath* and *Run just this toolpath* (the latter bounded to that section), each preceded by the `G90 G94` / `G17` / `G21` prolog.

The new `FusionFolderLoader` is a faithful C# port of the add-in's combine / strip / rapid-restore rules; `FolderPicker` uses the modern `IFileOpenDialog` (`FOS_PICKFOLDERS`) for folder selection.

`GCodeBlock` gains a `Section` tag for outline grouping and an optional `AddLineNumbers` gate; `GrblViewModel` gains `IsFolderView` and a one-shot `RunToBlock` end-bound consumed by `CycleStart`.

Companion Fusion 360 add-in (`ioSenderBatchPost`): a standalone add-in (no external framework) that posts every operation in every setup to its own `<seq>_<name>_T<tool>.nc` file in a chosen folder — the input for Load Folder. Combining, tool-change insertion and rapid restoration are intentionally left to ioSender, so the add-in only posts. A post-processor dropdown (default `grbl.cps`) selects which post Fusion runs per operation. Ships with `install-windows.ps1` / `install-macos.sh` and a README. Fusion auto-discovers it once copied, but it must be enabled once via *Scripts and Add-Ins*.

Tool table for the simulator: the add-in also writes a `0_ToolTable.nc` carrying `(STOCK X= Y= Z=)` and `(TOOL T= D= TYPE= [A=])` comment lines — the stock box size and each tool's diameter and shape (`FLAT`/`BALL`/`VBIT`, with the V-bit included angle) — read from the setup's stock parameters and tool definitions. Load Folder detects this file (name matches `tool table`), loads it first with comments preserved, and pushes the leading `(STOCK …)`/`(TOOL …)` lines to a connected grblHAL simulator, which uses them for its 3D material-removal carve (real controllers ignore the comments). A dialog checkbox toggles it; the output matches the SRWCommands PostProcess add-in byte-for-byte. See `TOOL_TABLE_FORMAT.md` in the simulator repo.

Note: the `Fusion Addin/` folder is an optional companion tool; the maintainer may prefer to host it separately. It can be dropped from the PR if so.

#7 Improve high-DPI / large-text-size layout across the UI

File	+	−
CNC Controls Camera/ … /ConfigControl.xaml	3	3
CNC Controls Lathe/ … /ConfigControl.xaml	6	6
CNC Controls Probing/ … /ProbingView.xaml	6	6
CNC Controls Probing/ … /ProbingView.xaml.cs	3	1
CNC Controls/ … /BasicConfigControl.xaml	4	4
CNC Controls/ … /CoordValueSetControl.xaml	6	6
CNC Controls/ … /FeedControl.xaml	5	5
CNC Controls/ … /GotoBaseControl.xaml	5	5
CNC Controls/ … /GotoControl.xaml	1	1
CNC Controls/ … /JogBaseControl.xaml	2	2
CNC Controls/ … /LEDControl.xaml	4	5
CNC Controls/ … /NumericField.xaml	6	3
CNC Controls/ … /NumericField.xaml.cs	3	1
CNC Controls/ … /OutlineBaseControl.xaml	8	4
CNC Controls/ … /OutlineControl.xaml	1	1
CNC Controls/ … /OutlineFlyout.xaml	1	1
CNC Controls/ … /OverrideControl.xaml	3	3
CNC Controls/ … /SpindleControl.xaml	8	8
CNC Controls/ … /StatusControl.xaml	3	3
CNC Controls/ … /StepperCalibrationWizard.xaml	2	2
CNC Controls/ … /StripGCodeConfigControl.xaml	15	9
CNC Controls/ … /ToolView.xaml	2	2
CNC Controls/ … /TrinamicView.xaml	1	1
CNC Controls/ … /WorkParametersControl.xaml	18	18
ioSender XL/ioSender XL/JobView.xaml	6	6
ioSender/ioSender/JobView.xaml	1	1

Makes the UI usable at 125–150% Windows text scaling by replacing fixed control heights/widths with auto-sizing plus `MinWidth`/`MinHeight`, so labels and fields grow with the font instead of clipping or overlapping. Changes are layout-only — no behavioural changes.

- **Shared controls:** the `NumericField` label column now auto-sizes (`MinWidth = ColonAt`) with a `SharedSizeGroup` so labels align in any container that sets `Grid.IsSharedSizeScope`. `CoordValueSetControl`, `StatusControl`, `LEDControl` (dropped a hard `MaxHeight`) and `OverrideControl` no longer pin `Height`/`Width`; the **Spindle** group switches its fixed `Width=250` to `MinWidth` so the RPM field is no longer clipped at larger text sizes.
- **grbl:Settings config controls** (Basic / StripGCode / Lathe / Camera) and both Stepper calibration tabs auto-size their rows and the Axis dropdown; the Probing view auto-sizes its left panel with shared-size label alignment.
- **Main window** (JobView XL + ioSender): right-panel grid / columns / panels auto-size instead of fixed 515 / 250 px, and each panel control (WorkParameters, Spindle, Coolant, Outline, Feed, Goto)

auto-sizes.

#8 Native Xbox controller support (+ Key Mappings editor polish)

File	+	−
CNC Core/CNC Core/XInput.cs	160	0
CNC Core/CNC Core/ControllerService.cs	186	0
CNC Core/CNC Core/ControllerMapper.cs	408	0
CNC Core/CNC Core/CNC Core.csproj	9	0
CNC Core/CNC Core/GrblViewModel.cs	16	0
CNC Core/CNC Core/KeypressHandler.cs	23	0
CNC Controls/CNC Controls/JogBaseControl.xaml.cs	5	0
CNC Controls/CNC Controls/KeyMapEditor.xaml	222	79
CNC Controls/CNC Controls/KeyMapEditor.xaml.cs	958	606
.gitattributes	3	2
ioSender XL/ioSender XL/App.config	6	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	259	0

Adds first-class game-controller support with **zero new dependencies** — the project uses no NuGet, so the controller is read via native **XInput** P/Invoke (`xinput1_4.dll`, falling back to `xinput9_1_0.dll`). A ~60 Hz `DispatcherTimer` poll layer runs in parallel to the keyboard handler.

- **Interop / service:** `XInput.cs` wraps the gamepad state and vibration structs; `ControllerService.cs` scans all four slots, edge-detects button down/up, and exposes Connected/Disconnected events, deadzone/normalize helpers and rumble.
- **Button map:** `ControllerMapper.cs` dispatches a default, fully re-mappable button map to machine actions (A = Cycle Start, B = Feed Hold, X = Spindle Stop, Y = Home, Back = Reset, Start = Unlock, D-pad = step-jog X/Y). Actions route through the same `GrblViewModel.ExecuteCommand` path as the on-screen buttons and are gated to a live connection in Idle/Jog/Tool states (with a status message when ignored). A short rumble confirms connection.
- **Analog jog:** the left stick proportionally jogs X/Y and the triggers jog Z, via throttled overlapping `$J=` moves (jog-cancel on release) so motion stays smooth; speed scales with stick deflection. Enable, left-stick deadzone, max-feed multiple of the Jog panel feed, and per-axis invert are user-configurable. The button map and analog settings persist to `ControllerMap.xml`.
- **Controller tab** in the Key Mappings editor: live press indicator, per-button action dropdowns (each button's default marked with a leading `*`), *Restore Defaults* (enabled only when modified), machine-direction-aware tooltips, and the analog-jog settings — machine dispatch is paused while the dialog is open. New `JogDistanceProvider`/`JogFeedProvider` hooks let controller jog match the on-screen Jog panel's current step and feed.

The same branch also carries the keyboard-tab polish (changed-binding highlight with per-row/group reset, *Clear* moved into the row menu, column sizing, and the grouped outline now keeps a section expanded while scrolling instead of collapsing it). The branch is fully LF-normalized, so the counts above are the real diff — no line-ending noise.

#9 Configurable main page + flyouts + jog panels (ioSender XL)

File	+	−
CNC Controls/CNC Controls/AppConfig.cs	141	20
CNC Controls/CNC Controls/AppConfigView.xaml	3	1
CNC Controls/CNC Controls/AppConfigView.xaml.cs	168	0
CNC Controls/CNC Controls/BasicConfigControl.xaml	2	0
CNC Controls/CNC Controls/CNC Controls.csproj	65	0
CNC Controls/CNC Controls/Converters.cs	17	0
CNC Controls/CNC Controls/JogBaseControl.xaml	2	2
CNC Controls/CNC Controls/JogBaseControl.xaml.cs	157	2
CNC Controls/CNC Controls/JogConfigControl.xaml	7	7
CNC Controls/CNC Controls/JogSlider.xaml	66	0
CNC Controls/CNC Controls/JogSlider.xaml.cs	51	0
CNC Controls/CNC Controls/JogUiConfigControl.xaml	18	9
CNC Controls/CNC Controls/KbdJogGridControl.xaml	46	0
CNC Controls/CNC Controls/KbdJogGridControl.xaml.cs	64	0
CNC Controls/CNC Controls/KeyboardJoggingControl.xaml	25	0
CNC Controls/CNC Controls/KeyboardJoggingControl.xaml.cs	34	0
CNC Controls/CNC Controls/LimitsControl.xaml	6	24
CNC Controls/CNC Controls/LimitsControl.xaml.cs	77	0
CNC Controls/CNC Controls/MachinePositionControl.xaml	69	0
CNC Controls/CNC Controls/MachinePositionControl.xaml.cs	21	0
CNC Controls/CNC Controls/MainPageEditor.xaml	108	0
CNC Controls/CNC Controls/MainPageEditor.xaml.cs	288	0
CNC Controls/CNC Controls/MainPanelRegistry.cs	111	0
CNC Controls/CNC Controls/NumericField.xaml	12	2
CNC Controls/CNC Controls/NumericField.xaml.cs	38	0
CNC Controls/CNC Controls/OffsetFlyout.xaml	41	0
CNC Controls/CNC Controls/OffsetFlyout.xaml.cs	133	0
CNC Controls/CNC Controls/PanelFlyout.xaml	43	0
CNC Controls/CNC Controls/PanelFlyout.xaml.cs	80	0
CNC Controls/CNC Controls/Properties/Settings.Designer.cs	36	0
CNC Controls/CNC Controls/Properties/Settings.settings	9	0
CNC Controls/CNC Controls/SidebarItem.cs	23	4
CNC Controls/CNC Controls/TabRegistry.cs	42	0
CNC Controls/CNC Controls/ToggleControl.xaml	20	2
CNC Controls/CNC Controls/UIJogGridControl.xaml	65	0
CNC Controls/CNC Controls/UIJogGridControl.xaml.cs	80	0
CNC Controls/CNC Controls/UIJoggingControl.xaml	44	0
CNC Controls/CNC Controls/UIJoggingControl.xaml.cs	29	0

CNC Core/CNC Core/KeypressHandler.cs	22	4
CNC Core/CNC Core/SerialStream.cs	1	1
ioSender XL/ioSender XL/JobView.xaml	23	14
ioSender XL/ioSender XL/JobView.xaml.cs	100	19
ioSender XL/ioSender XL/MainWindow.xaml	3	3
ioSender XL/ioSender XL/MainWindow.xaml.cs	123	5
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	322	0

ioSender XL only. Makes the crowded right-hand panel area configurable instead of fixed. A new **Edit Main Page** dialog (beside *Edit Key Bindings*) is a shuttle editor with three buckets — *Available*, *Main page* (up to six slots), and *Flyouts* — so each named panel can be placed on the main page or as a pinnable right-edge flyout, or left off entirely to reclaim space. Nothing is auto-assigned.

- **Flyouts** share a flush, tab-height border and distribute their content vertically; each has a pin (stays open when another flyout opens, and persists across launches) and a close button.
- **Per-offset flyouts** (G28 / G30 / G54..G59.3 / G92) are tiny *Go* buttons (tooltip shows the live coordinates; G28/G30 also get a *Set* = `G28.1/G30.1`) so several can be pinned beside the jog panel.
- **Jog panels** — UI jogging (distance + speed) and keyboard default-speed — become assignable panels rendered as compact sliders with a read-only value; Settings:App stays the editor. The jog-arrow radios are hidden only when the UI jog panel is assigned.
- **Settings:App** gains opt-in autosave (with a discard prompt) and a *Reset to default* context-menu entry on numeric fields; the old *Edit Key Mappings* button is renamed **Edit Key Bindings**.

Layout choices persist (panels / flyouts / pins serialized as CSV to sidestep the `XmlSerializer` list-append quirk). Also includes robustness fixes: a resilient `SerialStream.OutCount` that no longer throws on a dropped port, and a deferred main-panel build so startup can't race the config load.

Extended: a fourth bucket places panels *left* of the 3D view (DRO, Program limits, Machine Position now in the registry) in a scroll region that never overlaps the fixed Signals/Status block. Per-offset *Set* now also handles G54–G59.3 (captures the current machine position as that coordinate system's origin via `G10 L2 P<n>`). The UI jog panel renders as a 2×4 distance/speed grid with green-click selection, a *Continuous* toggle and a *Remember distance & speed across restarts* option; *Link distance and speed to UI jogging* makes keyboard jogging follow the UI distance *and* speed (hiding the Keyboard Jogging panel). Coolant toggles render as a self-contained red/green pill (no theme dependency). A new **Tabs** tab in the editor reorders / hides the main TabControl tabs (Settings:App protected), applied on startup via `TabRegistry` / `Config.Tabs`. The *Restart* button moved to Settings:App (next to *Edit Main Page*), enabled only after a layout/tab change.

Panel refinements: the Machine Position panel shows MPos as a live `work + WorkPositionOffset` sum (tracking the DRO notifications); the limits panel stays visible and shows the machine soft-limit envelope from `$13x/$23` when no program is loaded (program bounding box when one is), retitling itself *Machine limits* / *Program limits* accordingly; the 2×4 jog grid keeps its green selection highlight using the retained discrete index (survives keyboard/continuous-mode flips); and the **Tabs** editor now also lists tabs that host an `IGrblConfigTab` (no `ViewType`) by falling back to the `TabItem` name.

#10 Macro manager + run-only flyout

File	+	−
CNC Controls/CNC Controls/MacroManagerDialog.xaml.cs	316	0
CNC Controls/CNC Controls/MacroManagerDialog.xaml	81	0
CNC Controls/CNC Controls/MacroExecuteControl.xaml.cs	19	8
CNC Controls/CNC Controls/AppConfig.cs	19	0
CNC Controls/CNC Controls/AppConfigView.xaml.cs	10	0
CNC Controls/CNC Controls/MacroExecuteControl.xaml	8	16
CNC Controls/CNC Controls/CNC Controls.csproj	7	0
CNC Core/CNC Core/GCode.cs	5	0
CNC Controls/CNC Controls/JobControl.xaml.cs	3	2
CNC Controls/CNC Controls/MacroToolbarControl.xaml.cs	3	0
CNC Controls/CNC Controls/AppConfigView.xaml	1	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	126	0

Adds an **Edit Macros** button to the Settings:App page opening a macro manager — a DataGrid with one row per macro, in the spirit of Fusion's *Scripts and Add-Ins* dialog — and reworks the on-screen macro flyout for running.

- **Name** and a **Prompt** (confirm-before-run) flag are edited in-line; an assignable **Key** column (F1–F12) sets the function key (keys are unique; *Create* auto-assigns the first free one); a **File** column shows the referenced file for `@<path>` macros (full path on hover), blank for inline.
- **Edit** edits the macro's definition (inline G-code, or the `@<path>` reference line) and reads it back; **View** opens what it points to — the referenced file for an `@<path>` macro (created if missing, to author a new one), otherwise the code. **Create**/**Delete** add and remove rows.
- The **macro flyout** is now run-only: a macro-name dropdown plus a **Run** button (the Edit button is gone — editing lives in the manager). It pre-selects the most recently run macro, persisted in `Config.LastMacroId` and updated from every run path (flyout, toolbar, F-keys), falling back to the first.

The macro's function key is decoupled from its `Id` via a new `Macro.FKey` field (legacy configs migrated on load: `Id n` keeps `Fn`). The `@<path>` *run-time* resolution — loading and running the referenced file, with directive processing — is provided by **#11**; this PR adds the manager/flyout support for authoring, viewing, and repointing references.

#11 Macro Processor Enhancements

File	+	−
CNC Controls/CNC Controls/MacroProcessor.cs	582	0
CNC Controls/CNC Controls/MacroToolbarControl.xaml.cs	1	1
CNC Controls/CNC Controls/MacroExecuteControl.xaml.cs	1	1
CNC Controls/CNC Controls/JobControl.xaml.cs	1	2
CNC Controls/CNC Controls/CNC Controls.csproj	1	0
CNC Core/CNC Core/GCodeParser.cs	31	12
CNC Core/CNC Core/NGCExpr.cs	22	0
CNC Core/CNC Core/Grbl.cs	11	0
CNC Core/CNC Core/GrblViewModel.cs	9	2

A sender-side **macro pre-processor** (`MacroProcessor.Run`) that interprets parenthesised directives before the remaining lines are streamed to the controller. All macro entry points — the top toolbar, the on-screen macro flyout, and the F-key handler — now route through it (previously the toolbar called `ExecuteMacro` directly and silently ignored every directive).

Directives, each a no-op the controller would treat as a comment, interpreted by the sender instead:

- **`(PREREQ, cond, …)`** — require machine state *before* anything streams; if unmet the macro aborts with a message and nothing is sent. Conditions: `homed`, `idle`, `tlo`, `noalarm`, `connected`; stored positions / work offsets `g28`, `g30`, `g92`, `g54`–`g59.3` ("set" if any axis offset is non-zero, read fresh from `$#`); and any other token is required to be a controller `$I` build option (`NEWOPT`, e.g. `EXPR`), matched exactly and case-sensitively. `homed` reads grblHAL's live `sys.homed.mask` from `$#` rather than the change-based status `H:` cache, which can stay stale after a position-loss alarm.
- **`(MBOX, [buttons,] message)`** — modal hold; `OK`/`OKCANCEL`/`YESNO`, Cancel/No aborts the rest.
- **`(WAITIDLE)`** — hold streaming until the controller finishes what was sent and returns to Idle — needed after a controller-side job such as `$F=<file>`, which acks immediately and drops sender input while it runs.
- **`(PROMPT param, default [, label])`** — collect input values in a single up-front dialog and bind them two ways: assign the globals on the controller (so `$F=` files can read them) and substitute the references in the streamed body.
- **`@<path>`** — a macro whose body is a single `@<path>` line is a reference to an external file: that file's current contents are loaded and run in its place, re-read on every run — so a macro can be developed by editing the file directly. The loaded contents go through the same directive pipeline. (#10 adds the manager/flyout support for authoring and repointing these references.)

Supporting core changes so macros stream cleanly to an NGC-capable controller, all in the spirit of "don't reject what the controller can evaluate": `ExecuteMacro` now inherits the controller's expression support (as file streaming already did) and forwards verbatim any line its own parser cannot handle; `GCodeParser` recognises **named O-word subroutine calls** (`O<name> CALL […]`) and forwards them to the controller, which resolves the named sub from its filesystem; and the `$# [HOME:…:mask]` homed-axes mask is parsed for the authoritative `homed` check.

#12 Onboarding tooltips

File	+	−
CNC Controls/CNC Controls/Converters.cs	50	0
CNC Controls/CNC Controls/StatusControl.xaml	3	0
CNC Controls/CNC Controls/StripGCodeConfigControl.xaml	10	5
CNC Controls/CNC Controls/JogUiConfigControl.xaml	10	9
CNC Controls/CNC Controls/JogConfigControl.xaml	7	7
CNC Controls/CNC Controls/BasicConfigControl.xaml	3	3
CNC Controls/CNC Controls/FileActionControl.xaml	4	4
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	196	0

Tooltips aimed at users new to CNC, where ioSender's UI is otherwise terse.

- **State field** — a context-aware tooltip explains the current controller state in plain language (Idle, Run, Hold, Check, Tool, Door, …). For an **alarm** it appends the controller's specific message — e.g. `ALARM 4: Probe fail …`, pulled from the `GrblAlarms` enumeration loaded at connect — plus how to recover, so a bare `ALARM:4` is self-explanatory instead of a code to look up. (A new `GrblStateToTooltipConverter` drives a `ToolTip` binding on the State box.)
- **Settings:App** — a “what it does and why it’s useful” tooltip on each application setting (the grbl settings page already self-documents in its right-hand pane, so it’s left alone). Keyboard jogging (fast/slow/step feed & distance, link-step-to-UI), UI jogging (the four distance and four speed presets, noting distance and speed are selected *independently*), and GCode command stripping (M6/M7/M8/G61-G64, for machines lacking the matching hardware). Also fixes the “Keep MDI focus” tooltip (it had been copy-pasted from the buffering option) and the circular “Send comments” one.

#13 Build documentation: Mega-XL upgrade notes + repeatable tool-change guide

File	+	−
docs/Mega-XL-Upgrades.html	840	0
docs/Mega-XL-Upgrades.pdf	bin	0
docs/Repeatable-Tool-Change.html	419	0
docs/Repeatable-Tool-Change.pdf	bin	0

Two self-contained HTML documents (each rendered to a companion PDF), added under a new `docs/` folder. Purely additive; touches no source.

- **Repeatable-Tool-Change** — a turnkey, novice-friendly tool-change guide built around a **G59.3 fixed toolsetter** and a `T<n>/M6` flow, with a top-down machine travel diagram (color-coded paths). Aimed at the goal of repeatable, hands-off tool changes for users new to CNC.
- **Mega-XL-Upgrades** — build notes for an upgraded Kickstarter v1.0 Mega V XL: frame/spindle/motion/probing/power sections, a back-of-enclosure two-panel harness model (hand-built SVG diagrams), a cable & wiring schedule, and an **interactive bill-of-materials worksheet** — an in-page editor (double-click to edit Qty / Unit / Description / Source, live extended & category subtotals & grand total, URL shortening, Save-to-download) plus a linked table of contents.

The Mega-XL notes are specific to one machine and may be of limited upstream interest — they can be hosted separately or dropped from the PR. The repeatable tool-change guide is more generally applicable. The associated controller macros (`tc.macro`, `probe_tfl.macro`) and the ATC provisioning UI are intentionally *not* in this PR — they remain with the macro work, pending firmware support.

#14 SD + LittleFS file browser & file manager on the SD Card tab

File	+	−
CNC Controls/CNC Controls/SDCardView.xaml.cs	505	111
CNC Controls/CNC Controls/GrblFilesystems.cs	142	0
tests/GrblFilesystemsTests.cs	83	0
CNC Core/CNC Core/TelnetStream.cs	54	14
tests/README.md	22	0
CNC Controls/CNC Controls/SDCardView.xaml	19	6
CNC Core/CNC Core/YModem.cs	14	4
CNC Controls/CNC Controls/CNC Controls.csproj	1	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	105	0

The SD Card tab listed only one filesystem (the SD card, or whatever was the current filesystem). On grblHAL builds that have both an SD card and a LittleFS store, the other filesystem — and any macros on it — were invisible.

This enumerates every mounted filesystem via `$FI` (`[FS:<name>@<path> size, free]`) and lists them together in one grid with a new **Location** column, plus a per-filesystem **free-space banner** above the list. The right-click menu is reworked from *Run* / *Download and run* / *Delete* into a small **file manager**, and file operations target the file's own filesystem — a bare name on the root volume (unchanged single-SD behaviour) or an absolute path on a sub-mount such as `/littlefs`.

- **File manager menu:** *View* / *Edit* / *Create* (new content, or from a local file) open the file in Notepad — Edit and Create write the result back to the controller when Notepad closes; *Load* and *Load and run* bring the program into ioSender; *Copy to* / *Move to* transfer a file to another mounted filesystem (submenus built live from the mounts); *Delete*. Actions are enabled only for a real selected file.
- **Empty filesystems are visible:** a mount with no files gets a placeholder row so it can still be selected as an upload / copy target; the listing's re-entrant `Load()` is guarded (it pumps the dispatcher while waiting on the controller, so an overlapping refresh could corrupt the shared table).
- **Correct attribution under nested mounts:** a root `$F` recurses into sub-mounts, so an SD-root listing also returned the `/littlefs/*` files; each result is now attributed to its deepest containing mount, so files are not duplicated under (and mis-pathed on) the parent filesystem.
- **Upload to any filesystem:** `UploadLocalFile` targets a chosen filesystem via `$CWD` (the FTP path is built from the explicit destination, not the possibly-stale `FsCwd`), and upload is allowed on a LittleFS-only controller (no SD card present, so no SD mount status to require).
- **Upload reliability:** `TelnetStream` now writes *synchronously* — overlapped `WriteAsync` calls silently dropped a YModem block's payload/CRC — and `PurgeQueue` clears the async-read buffer instead of racing a second synchronous read on the same stream (which lost per-block ACKs, producing 0-byte files and multi-minute uploads); plus a bounded connect timeout so a reconnect retries quickly. YModem per-block ACK timeouts are raised (10 s open / 5 s block) to ride out flash-filesystem GC stalls, and blocks are sent in **128-byte (SOH)** packets rather than 1024-byte (STX): grblHAL's YModem receive ring is 1024 bytes (usable 1023), so a full 1029-byte STX packet can't fit — over telnet it arrives in a TCP burst faster than the firmware drains it and overflows,

stalling the transfer (a one-block file uploaded, larger ones hung). 128-byte blocks fit with margin on any transport.

- **Backward compatible:** builds that don't implement `$FI` (or report nothing mounted, `error:65`) return no `[FS:…]` lines, and the view falls back to the original single-filesystem mount + `$F` listing untouched. An SD card with files in its root works exactly as before, now also showing free space.
- **New `GrblFilesystems.cs`** holds pure, dependency-free helpers (mount-line parser, human-readable size, path qualification, free-space summary) and the `FsMount` model; the live `$FI`/`$F` querying lives in `GrblSDCard`.
- **Unit tested:** `tests/GrblFilesystemsTests.cs` exercises the pure helpers (29 assertions, run via the Framework `csc` — see `tests/README.md`; the app has no test project).

Builds clean (CNC Controls and the full ioSender solution). Also the groundwork (and the reliable LittleFS upload path) that the ATC macro-provisioning work reuses to target the `$FI`-discovered LittleFS path instead of a hardcoded one.

#15 USB / Ethernet / simulator connection support + Connect menu

File	+	−
CNC Controls/CNC Controls/SimulatorManager.cs	474	0
ioSender XL/ioSender XL/MainWindow.xaml.cs	208	14
CNC Controls/CNC Controls/AppConfig.cs	165	28
CNC Controls/CNC Controls/BasicConfigControl.xaml	1	0
CNC Controls/CNC Controls/PortDialog.xaml.cs	56	2
CNC Controls/CNC Controls/PortDialog.xaml	43	8
ioSender XL/ioSender XL/MainWindow.xaml	39	2
CNC Controls/CNC Controls/GrblConfigControl.xaml.cs	22	0
simulator/README.md	18	0
CNC Core/CNC Core/TelnetStream.cs	14	3
CNC Controls/CNC Controls/JobControl.xaml.cs	14	2
CNC Controls/CNC Controls/UIUtils.cs	12	4
ioSender XL/ioSender XL/ioSender XL.csproj	12	0
CNC Core/CNC Core/HelperClasses.cs	8	2
ioSender XL/ioSender XL/JobView.xaml.cs	18	2
CNC Core/CNC Core/GrblViewModel.cs	7	0
readme.md	2	0
CNC Controls/CNC Controls/CNC Controls.csproj	2	0
CNC Controls/CNC Controls/Widget.cs	1	1
.gitignore	5	0
CNC Controls/CNC Controls/GrblConfigControl.xaml	1	0
simulator/sim-build.json	1	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	28	0

Adds first-class connection setup to the sender: pick USB (serial), Ethernet (`host:port`, now accepting hostnames as well as IPv4), or a local grblHAL **simulator** from a tab in the connection dialog, and (re)connect at any time from a menu. *(Updated 2026-06-20: the simulator is now run standalone and connected to over a port; the launch/download/My-Machine dialog was stripped out, and connection-loss + comment-forwarding fixes were folded in.)*

- **Simulator tab — connect-only:** the simulator now runs *standalone* — the user launches `grblHAL_sim.exe` directly (it opens its own 3D view + config window and listens on a TCP port), and ioSender simply connects to `127.0.0.1:<port>`. The tab is reduced to a single *Port* field ("Launch grblHAL_sim.exe first, then connect"); ioSender no longer launches, downloads or manages the simulator process.
- **Connect / Reconnect:** a `File > Connect…` item opens the dialog; while connected it becomes *Reconnect…* (disconnects, then re-opens the dialog so a different target can be chosen). The app now stays open when started without a connection instead of exiting, so the menu is reachable.
- **Remembered network host:** the dialog's network tab defaults to the most recent IP that connected successfully (persisted as `Config.NetworkHost`) rather than always the mDNS name `grblHAL.local`. It starts empty; on a serial connection it is seeded from the controller's `$I-`

reported IP if still unset — so after one USB session a later reconnect already offers the controller's real address. Empty falls back to `grblHAL.local`; the bundled simulator (`127.0.0.1`) is excluded.

- **Prefer network connection (opt-in):** a Settings:App checkbox (`Config.PreferNetwork`, default off). After a serial/USB connection whose `$I` reports an IP, ioSender probes `<ip>:23` and, if it answers, automatically switches the live link to the network — status line: *"Connection migrated to network (ip:23)"* — falling back to the serial port if the network connect fails. The probe runs off the UI thread; the switch is deferred and guarded against re-entry, and only fires from a serial link with a known IP.
- **Target status & cues:** the status bar shows the current connection target (`GrblViewModel.ConnectionTarget`), green when connected and red when not. The connect dialog opens on the tab matching the active target (Serial / Network / Simulator, disambiguated via the `StartSimulator` flag) and green-bolds that header; the XL window background tints pale yellow while connected to the **simulator** (restored to grey on a real target or when disconnected), so a virtual machine is unmistakable.
- **"Not connected" on a dropped socket:** when the link drops (e.g. the simulator window is closed) the reconnect loop keeps the target name set, so the target box would otherwise still read connected. An `IsConnectionLost` trigger now turns the XL target box red and shows *Not connected* until the connection is re-established.
- **Stop the job on disconnect:** if the controller connection drops *during a running job*, the **client-side** stream is torn down (job state reset to idle) so the UI no longer hangs waiting on `ok` responses that will never arrive — previously this was handled cleanly only while idle. This is a Stop, not a feed-hold, and because the link is already gone it cannot command the controller; what the machine does physically on comms loss is up to the firmware.
- **Always send comments to the simulator:** g-code comments are forwarded to the simulator regardless of the app's *Send comments* setting — the simulator reads them for tool/stock metadata (`(TOOL …)`, `(STOCK …)`) that drives its 3D view and material-removal carve.
- **Startup ordering:** `AppConfig.SetupAndOpen` is split into `LoadConfig` (run synchronously at construction, so config is available before any control's `Loaded` handler reads it) and `OpenConnection` (deferred to `ApplicationIdle` so the main window paints before the — possibly blocking — connection dialog). Transport is chosen by target string (`COMx` vs `host:port`) rather than a leading-digit IPv4 check.
- **Copy current settings to the simulator:** a *Copy to simulator* button on the grbl:Settings tab (next to Save) mirrors the live controller settings into the bundled simulator's "My Machine" EEPROM via `SimulatorManager.BuildMyMachineEeprom`, so a later simulator connection boots with this machine's configuration.

Builds clean (CNC Controls and the full ioSender XL + classic solutions) off `master`. The "My Machine" EEPROM builder is now wired to the *Copy to simulator* button (above); the `SimulatorManager` launch / web-builder-download machinery is no longer wired to the (now connect-only) dialog — a candidate for removal in a follow-up. No simulator binary is committed; see `simulator/README.md`.

#16 Validate controller (check-mode g-code conformance test)

File	+	−
CNC Controls/CNC Controls/ValidateProcessor.cs	1091	0
CNC GCodeViewer/CNC GCodeViewer/Renderer.xaml.cs	81	16
ioSender XL/ioSender XL/MainWindow.xaml.cs	3	21
CNC Controls/CNC Controls/CNC Controls.csproj	1	0
CNC Controls/CNC Controls/LibStrings.xaml	3	0
ioSender XL/ioSender XL/MainWindow.xaml	1	1
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	21	0
Locale/*/csv/ioSender.resources.*.csv (7 locales)	7	0

Replaces the placeholder *Validate controller* action (#15) with a real, in-app conformance test: it streams a g-code program tailored to the connected controller's reported capabilities and reports which commands it accepts — without moving the machine.

- **Capability-tailored test:** generated from `$I` build options, axis count, `$32` mode, tool count, aux-I/O counts and the work coordinate systems present, so a feature the firmware was never built with is not tested (and cannot be reported as a false failure).
- **Check mode, no motion:** streamed in `$C` check mode and parsed only — the machine never moves, so homing and a clear envelope are not required. One feature per line, so an `error:N` pinpoints exactly which command the controller rejected.
- **Robust against grbl's check-mode behaviour:** a check-mode error *latches* the parser (every later line then errors), so the run recovers on each error — reset, re-enter check mode (unlocking first if the reset re-alarmed a homed machine), re-apply the modal set-up — then resumes, always re-confirming check mode before sending another line so a test line can never reach the controller as real motion.
- **Non-destructive:** check mode *commits* `G10`/`G92` writes, so the run snapshots and restores the work offsets / tool table and avoids the un-restorable writes (`G28.1`/`G30.1`, and `G92` when an offset is active). Machine settings (`$$`) are never written.
- **Progress & results:** a small non-modal panel shows a live pass/fail tally and the current test as it streams; on completion a *View Summary* button opens the full report (grouped by category, error code/message per failure, the pre-run parameter snapshot, Copy-to-clipboard). The controller is left in `Idle` with check mode off.
- **3D visualization:** the passed bounded-motion features are assembled into a runnable program (absolute `G90` moves at a fast feed) and loaded into the buffer, so the user can press Cycle Start and watch the toolpath run in the 3D view via the normal job path. The viewer also gains an *always-active machine scene* (work envelope, axes and a live tool cone shown when homed even with no program loaded), plus fixes to the tool/executed-trail animation subscription and a bounded executed-trail advance (an out-of-range block was an unbounded spin).

Right-click the **Target** status item → *Validate controller*. Builds clean; the only failures on a stock grblHAL are commands it genuinely does not implement (e.g. `G90.1`, `G61.1`, `G64`, `M52`; `M56` needs parking-override enabled).

#17 Restore main window position across launches

File	+	−
`ioSender XL/ioSender XL/MainWindow.xaml.cs`	58	7
`CNC Controls/CNC Controls/AppConfig.cs`	2	0

The "Restore last window size on startup" option restored only the window *size*; the position was discarded (clamped to the top-left). This persists the position too.

- **Position persisted:** new `WindowLeft`/`WindowTop` settings, saved on close (using `RestoreBounds` when maximized so the un-maximized rectangle is stored) and restored on startup — but only when the saved rectangle lands on a currently-connected monitor (checked against the whole virtual desktop), so a window saved on a since-removed screen can't open off-screen.
- **Takes effect immediately:** the live "save" flag was only set at startup, so toggling the option mid-session did nothing until a restart. It now updates live and captures + persists the current placement the instant the option is enabled.

Builds clean. The size/position restore builds on the connection support's `CompleteStartup`; the underlying `KeepWindowSize` option is from the ioSender baseline.

#18 grbl:Settings — live edits, change highlight, revert & snapshots

File	+	−
CNC Core/CNC Core/Grbl.cs	356	15
CNC Controls/CNC Controls/GrblConfigControl.xaml	49	6
CNC Controls/CNC Controls/GrblConfigControl.xaml.cs	39	8
CNC Controls/CNC Controls/Widget.cs	25	14
CNC Controls/CNC Controls/RestorePointDialog.xaml	41	0
CNC Controls/CNC Controls/RestorePointDialog.xaml.cs	65	0
CNC Controls/CNC Controls/CNC Controls.csproj	7	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	77	0

Orca/PrusaSlicer-style indication of which controller $ settings you have changed, on the grbl:Settings **Basic** tab, plus live editing feedback, granular revert and automatic per-Save restore points. Applies to both the grblHAL settings tree and the classic-grbl grid.

- **Live edits in the left-hand list:** edits stage in a pending-changes map (`GrblSettings.PendingEdits`) that the list reads via a new `EditValue`, so the value and its highlight update *immediately* without mutating the controller value. Checkboxes/radios commit on change; text/numeric editors commit on lost focus. Pending edits are flushed to `Value` and cleared on Save; discarded on Reload/Restore.
- **Two highlight states:** **orange/bold** = unsaved edit (`IsEdited`); blue = saved to the controller this session but still different from the session-start value (`IsSavedModified`). Groups roll up the same way.
- **Revert from a right-click menu:** *Revert to value at startup* on a setting, and *Revert all in group to value at startup* on a group node (staged as a pending edit, written on next Save).
- **Snapshot on Save + Restore picker:** every Save writes a timestamped restore point (full settings dump in the Backup format, led by a `; changes (N):` old→new comment block), per controller identity, keeping the last 20. The **Restore** button opens a picker listing restore points newest-first with a changed-count column and a per-row tooltip showing exactly what each Save changed; *Browse…* falls back to any backup file.
- **Baselines, cleanly separated:** `GrblSettingDetails` tracks the current value, the frozen *startup* value (highlight) and the *loaded* controller value (drives `IsDirty`). Making `IsDirty` mean "differs from the controller" rather than "changed" fixes a spurious "settings changed, save now?" prompt when reverting an edit back to the controller value.
- **Bitfield flag tooltip:** `BITFIELD`/`XBITFIELD` settings still display numerically, but hovering the value shows the set flags decoded to their labels (one per line, `N/A` entries skipped) via `GrblSettingDetails.BitfieldLabels`.

Builds clean off `master` (CNC Controls and the full ioSender solution); verified against the bundled grblHAL simulator. Per-setting "revert to factory default" is intentionally deferred — grblHAL exposes no per-setting default (only the global $RST=$ reset-all), so factory revert would need a future firmware-side snapshot of as-flashed values.

#19 Console — inline terminal-style command prompt

File	+	−
CNC Controls/CNC Controls/ConsoleControl.xaml.cs	117	0
CNC Controls/CNC Controls/ConsoleControl.xaml	28	4
CNC Core/CNC Core/GrblViewModel.cs	20	0
CNC Core/CNC Core/KeypressHandler.cs	7	0
CNC Controls/CNC Controls/MDIControl.xaml.cs	5	16
CNC Controls/CNC Controls/MDIControl.xaml	1	1
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	42	0

Makes the **Console** tab read like a traditional terminal by adding a command input line at the bottom, while keeping the global single-line MDI strip for the other tabs (3D View, Program). A natural answer to "why is MDI a separate strip instead of a console prompt?" — now it can be both.

- **Reuses the MDI plumbing:** Enter routes the line through the existing `MDICommand`, which already echoes the command into the console's `ResponseLog` — so the typed command and its response read inline like a shell. No `JobView` changes.
- **Shared command history:** a single `GrblViewModel.MDIHistory` (newest first) is fed by both the MDI strip and the console prompt and surfaced in both — the strip's dropdown binds to it and the prompt recalls it with Up/Down (Esc clears). A command entered in either place is recallable in the other. (The view-model `CommandLog` is not populated for these entries, so it is not used.)
- **Same guards, auto-focus:** the prompt is locked out while a job runs (same as the MDI strip) and grabs focus when the console becomes visible (tab switch or the pop-out console window) so it's ready to type into.
- **Pop-out for free:** the prompt is self-contained in `ConsoleControl`, so the detachable console window (which hosts the same control) gets it too.
- **Output context menu:** right-click the response log for *Clear all*, *Clear up to here* (drops the clicked line and everything above it), and *Save…* (writes the in-memory log to a text file).
- **No jog conflict:** the keyboard-jog handler jogs whenever a key is *mapped* to jog (Up/Down are by default) *before* its per-key text-box guard, so the prompt's arrows would otherwise move the machine. `ProcessKeypress` now forces no-jog when the focused element is a `TextBox` tagged `"MDI"`, and the prompt is tagged accordingly — uniformly exempting both the MDI strip's edit box and the console prompt.

Builds clean off `master` (CNC Controls and the full ioSender solution); verified against the bundled grblHAL simulator: commands echo + execute, auto-focus, history recall (Up/Down step through prior commands), and an A/B test confirming Up jogs the DRO when a non-input element is focused but does not when the console prompt is focused. Keys are handled in `PreviewKeyDown` so the arrows reach the prompt rather than being consumed by the TextBox.

#20 3D view — Ctrl+click to jog + per-axis orientation

File	+	−
CNC GCodeViewer/CNC GCodeViewer/Renderer.xaml.cs	164	52
CNC Core/CNC Core/Grbl.cs	26	1
CNC GCodeViewer/CNC GCodeViewer/RenderControl.xaml	11	7
CNC Core/CNC Core/GCodeEmulator.cs	3	0
CNC Controls/CNC Controls/AppConfig.cs	2	0

Ctrl+left-click in the 3D or top-down (XY) view jogs the machine to the clicked XY position — a quick coarse-positioning aid that complements the jog buttons (park over a corner to set work zero, check alignment, touch-screen rigs).

- **Safe by construction:** the move always retracts **Z to the top of travel first**, then rapids to the new XY, so the traverse can never plough through stock or fixtures. Only X/Y move; plain left-drag still rotates the camera.
- **How it maps:** the click ray is intersected with the work Z=0 plane (HelixToolkit `UnProject`) to recover an XY target, converted to machine coordinates (`MPos = WPos + WCO`), clamped to the soft-limit travel exactly as the jog limiter does, and emitted as two queued `$J=G53` jogs at the controller's max rate (`$110`–`$112`) so it moves at rapid speed.
- **Guarded:** requires homed + idle + max travel set + soft limits (`$20`) on, with a status-bar reason when a prerequisite is missing. Gated by a `GCodeViewerConfig.ClickToJog` setting (default on).
- **Orientation fix (prerequisite):** the rendered machine frame — work envelope, floor grid and crosshair — now honors the `$23` homing-direction mask + force-set-origin *per axis*, so the view matches any home corner (e.g. top-back-left, X+/Y−/Z−) instead of assuming +X/+Y. Standard +X/+Y/−Z machines render identically to before. `GrblInfo.ForceSetOrigin` now reads the live `$22` bit 3 rather than the `$I` `OPT` 'Z' flag (absent on some builds, e.g. the simulator), so the orientation is correct even when the option string omits it. The camera bounding box in the always-on machine scene is derived from the same per-axis envelope, fixing a mirrored home corner in the 3D view.
- **Toolbar tidy-up:** the 3D-view toolbar tooltips now show on disabled buttons and are reworded for consistency, the *save view* button is no longer needlessly gated, and the *show job envelope* button enables only with a file loaded. A null-tokens guard in `GCodeEmulator` avoids a crash when the machine scene renders with no program loaded.

Builds clean off `master` (CNC GCodeViewer chain). Verified on the author's machine: Ctrl+click retracts Z then rapids to the clicked XY at the configured max rate, and the envelope/grid orient to the real home corner.

#21 grbl:Settings — Machine Setup Wizard

File	+	−
CNC Controls/CNC Controls/MachineSetupWizard.xaml.cs	795	0
CNC Controls/CNC Controls/MachineSetupWizard.xaml	278	0
CNC Controls/CNC Controls/MachineCatalog.cs	194	0
CNC Controls/CNC Controls/GrblConfigView.xaml.cs	30	0
CNC Controls/CNC Controls/CNC Controls.csproj	8	0
CNC Controls/CNC Controls/GrblConfigView.xaml	4	0
CNC Controls/CNC Controls/AppConfig.cs	2	0
CNC Controls/CNC Controls/IGrblConfigTab.cs	2	1
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	427	0

A guided, first-run **Machine Setup Wizard** tab in grbl:Settings that configures the machine-description settings a fresh controller needs — so a new machine can be described without hunting through the $ list. Auto-selected when the machine looks unconfigured (no remembered machine, or $130–$132 all zero).

- **Single page, nothing written until applied:** the whole machine description is laid out on one page — machine selector, a clickable home-position picture, a per-axis table, and the remaining homing & limits questions — so it reads at a glance instead of an eight-step click-through. *Apply* writes the changes via `GrblSettings.Save()`.
- **Apply only when there's something to write:** Apply is disabled while the wizard's values already match the controller; the pending-change set recomputes on every edit (machine pick, field, reload). Numeric settings are compared by value, so a formatting-only difference (`1` vs `1.000`) is not flagged as a change. Hovering Apply lists the exact pending writes (`$id name: old → new`).
- **Out of the way:** sits as the *last* grbl:Settings tab (rarely needed after initial commissioning), so Basic settings is the default tab.
- **Machine catalog:** a built-in catalog (`MachineCatalog.cs`) of common hobby machines as manufacturer → product → model drop-downs (or a custom X×Y), each carrying a sensible home corner and force-set-origin default; the first entry is a generic XYZ machine with limit switches. Picking a model pre-fills the page, which the user then confirms or tweaks.
- **Axis table:** an editable per-axis grid — travel limit ($130–$132), max rate ($110–$112), steps/mm ($100–$102), invert direction ($3) and normally-closed limit ($5) — edited in place.
- **Home corner:** a clickable isometric box — click the corner where the machine homes — sets the X/Y homing directions ($23) and auto-enables force-set-origin ($22 bit 3) on grblHAL; Z is fixed to the top of the gantry. This is the same convention that orients the 3D view (#20). The axis table also carries a per-axis **Home min ($23)** checkbox kept in sync with the picture, so the homing direction is visible/editable per axis (Z's box is locked to "top").
- **Homing & limits:** hard limits ($21) default on, with the remaining global questions as a short list and the explanatory help moved into per-input tooltips. A right-justified *Firmware Info* button shows the live $I build output in a non-modal dialog.
- **Remembered & forgettable:** the chosen machine is saved (`AppConfig.LastMachine`) and restored next launch — keeping the live controller values over the catalog preset on restore — and a *Forget machine* button clears it so the wizard reappears next time.

- **Correct write order & honest errors:** grblHAL rejects soft limits (`$20=1`) unless homing is enabled (`error:10`). Apply now enables homing by the user's intent (writing `$22` even on a fresh controller where it is currently off, instead of gating on the live state) and writes soft limits in a second pass — after `$22` — since `GrblSettings.Save()` writes in ascending id order. Soft limits are only requested when homing will be on. On failure it lists the exact rejected `$` settings and the controller's reason rather than a generic “failed to write settings”.

Builds clean off `master` (CNC Controls and the full ioSender solution). UI verified by the author against a live controller and the bundled simulator; interactive `$3` direction-verify and homing-cycle-order editing were intentionally left out of this first cut. The first-run gate that forces the wizard to the foreground until a machine is applied — and its skip-when-connected-to-the-simulator behaviour — depend on connection-side plumbing and ride along in the integration branch rather than this stand-alone PR. The “copy this machine into the simulator’s My Machine profile” action now lives on the grbl:Settings tab (#15).

#22 Keyboard jog — keep active when Job-view focus drifts

File	+	−
ioSender XL/ioSender XL/MainWindow.xaml.cs	19	0
ioSender XL/ioSender XL/JobView.xaml.cs	4	1

Keyboard jogging only fired through `JobView.OnPreviewKeyDown`, which needs keyboard focus inside the Job view's tree. Interacting with a flyout, side panel or menu moves focus out, so the arrow keys stopped jogging until the view was re-focused (e.g. by switching tabs). This keeps jog alive on the Job page wherever focus has drifted.

- **Window-level forwarding:** the existing `MainWindow` tunneling `PreviewKeyDown` (always fires) forwards jog keys to `JobView.ProcessKeyPreview` when the Job view is the current view but is not keyboard-focused. The focused case is unchanged — `JobView` still handles it, including its MDI/DRO/spindle/work-params jog gates.
- **Skips text input:** forwarding is suppressed when focus is in any `TextBoxBase`, so typing in the MDI strip or console prompt never jogs.
- **Clean stop:** a mirrored `PreviewKeyUp` forwards the key-up too, so a continuous jog started while the view was unfocused still receives its release and stops (no text-input guard here — an in-progress jog must always cancel).

Builds clean off `master` (the full ioSender solution). Two-file change reusing the console-shortcut `PreviewKeyDown` hook already on `master`; `ProcessKeyPreview` is promoted to `public` so the window can forward into it.

#23 Go To — optional safe-Z (lift, traverse, descend)

File	+	−
CNC Controls/CNC Controls/GotoBaseControl.xaml.cs	34	8
CNC Controls/CNC Controls/AppConfig.cs	5	0
CNC Controls/CNC Controls/BasicConfigControl.xaml	1	0
Locale/*/csv/CNC.Controls.WPF.resources.*.csv (7 locales)	14	0

A **Go To** a work-coordinate origin (G54–G59.3), `G28` or `G30` was a single coordinated rapid — a straight diagonal in machine space that can drag Z through clamps, fixtures or tall stock on the way. This adds an optional safe-Z decomposition so the traverse always happens clear of anything standing up in Z.

- **Three-phase move:** with *Safe Z on Go To* enabled, each Go To becomes `G53 G0 Z0` (retract to machine top), then `G53 G0 X…Y…` (traverse at the safe height), then `G53 G0 Z…` (descend to the target). Machine `Z0` is the homed top — already top-minus-pull-off with force-set-origin — so it never re-trips the Z limit, and it needs no configured clearance height (it clears anything that physically fits under the gantry).
- **All Go To targets:** applies uniformly to the selected work origin and to G28/G30, read from the same `$#`-populated coordinate table the work-origin button already used (so G28/G30 get true X/Y-then-Z, not a firmware diagonal). Rotary axes travel with X/Y; only Z is separated.
- **Guarded, with a clean fallback:** gated on the *Safe Z on Go To* option (Settings:App → Main, default on), the machine being **homed** (machine coordinates valid) and **soft limits** (`$20`) on so the three rapids are envelope-checked. If any is missing it falls back to the original single coordinated move — no behaviour change for unhomed/soft-limits-off setups.

Builds clean off `master` (both apps). The setting lives in the Base app config so it shows in the always-present Main settings panel regardless of jog mode; grbl's junction deviation (~0.01 mm) means the corners are effectively full stops, so no dwell is needed between phases.

#24 comms — fix telnet/websocket UI deadlock

File	+	−
`CNC Core/CNC Core/TelnetStream.cs`	17	6
`CNC Core/CNC Core/WebsocketStream.cs`	12	4

Over a telnet or websocket connection the UI could freeze whenever a modal message box (e.g. a macro `M0`/MBOX hold) was open while polling continued — a lock-ordering **deadlock** between the read callback and the UI thread.

- **The deadlock:** `ReadComplete` ran a synchronous `Dispatcher.Invoke(DataReceived)` *while holding* `lock(input)`. The UI thread's `PurgeQueue` also takes `lock(input)`, so once the dispatcher was busy (a modal pumping its own message loop) the read callback waited on the UI thread while the UI thread waited on the lock — a classic cyclic wait that hung the app.
- **The fix:** extract the complete replies from the buffer *under* the lock, then release it *before* dispatching them to `DataReceived`. The serial and Eltima streams already dispatched after releasing the lock (via `BeginInvoke`); this brings telnet/websocket in line.

Builds clean off upstream `master` (2.0.47); a pure comms-layer fix, no UI or feature change. Surfaced while testing the ATC Start Job sequence (#25), whose install-probe MBOX hold kept an active telnet poll — exactly the deadlock window.

#25 ATC tool-change macro provisioning (Install ATC + Start Job)

File	+	−
CNC Controls/CNC Controls/AtcMacros.cs	369	0
CNC Controls/CNC Controls/SDCardView.xaml.cs	80	3
CNC Controls/CNC Controls/SDCardView.xaml	4	1
CNC Controls/CNC Controls/CNC Controls.csproj	11	0
CNC Core/CNC Core/Grbl.cs	14	1
CNC Core/CNC Core/YModem.cs	14	5
ioSender XL/ioSender XL/JobView.xaml.cs	6	0
ioSender/ioSender/JobView.xaml.cs	6	0
macros/tc.macro	105	0
macros/probe_tfl.macro	140	0
macros/start_job.macro	52	0
macros/cal.macro	3	0

Turnkey **repeatable tool change** for ATC controllers: ioSender ships the controller-side tool-change macro set and installs it to the controller's persistent **LittleFS**, then seeds an ioSender-side "Start Job" macro that drives the home → set-WCS → reference-toolsetter → run sequence. The novice only sets `G28/G30/G59.3` and supplies a 3D probe + toolsetter.

- **Embedded macros + provisioning:** `AtcMacros.cs` carries `cal/probe_tfl/tc.macro` as embedded resources and uploads them to `/littlefs` over **YModem** (absolute `/littlefs/<name>` path so they land on the right filesystem even with an SD card mounted at root; a `atc.sum` SHA-256 sidecar detects drift / re-installs).
- **Explicit "Install ATC macros" button** on the SD Card tab is the *only* provisioning trigger — user-initiated, re-verifying the filesystem each click. Auto-on-connect/auto-on-show were removed after proving fragile against controller-I/O timing (could wedge the link or hang on close).
- **`$I` capability parsing** (`Grbl.cs`): `ATC=1` → `HasATC`, `ATC=0` → `AtcMacrosRequired` (an ATC is configured but its `tc.macro` is missing — prompt to install + reboot), `YM` → `HasYModem`. A 2-arg `YModem.Upload(path, remoteName)` overload targets an absolute remote path.
- **Start Job macro** (both apps): seeded on connect to an ATC controller (`@start_job.macro`, run line-by-line through the macro pre-processor). The controller-side macros it `CALL`s stay on LittleFS, resolved by `O<cal> CALL` / `M6`. The Start Job go-to `G30/G28` moves are written as an explicit safe-Z sequence (lift to machine top, traverse X/Y, descend) rather than a bare diagonal rapid; the Z probes descend to `−140` (below the spoilboard, within `$132`).
- **Embed-on-Build:** `CNC Controls.csproj` disables VS's fast up-to-date check so edits to the external `..\..\macros\*.macro` embedded resources are actually re-embedded on a normal Build (the fast check ignores external files and would otherwise ship a stale copy).

Builds clean in Release for both apps; the full Start Job sequence was verified end-to-end on hardware (2026-06-18). Kept self-contained on PR 14 by seeding the Start Job macro without an auto function-key (that nicety needs PR 10's `Macro.FKey`); assign one in the Macro manager if wanted. The `O<cal> CALL` error:81 fix lives in PR 11 (label/keyword gap-close) plus a firmware-side macro-search-path fix.

#26 Offsets — editable G28/G30 (Set rapids & stores)

File	+	−
`CNC Controls/CNC Controls/OffsetView.xaml.cs`	21	2

The predefined offsets `G28` and `G30` were read-only in the Offsets view — you could only store the *current* machine position. This makes them **editable**: type a target and *Set* now honours it.

- **Set rapids, then stores:** because `G28.1`/`G30.1` can only capture the live position, *Set* first rapids the machine to the edited target with `G53 G0`, then stores it with `G28.1`/`G30.1`. A confirmation prompt spells out the move first (it physically moves the machine); a pending unit change is honoured around the move.
- **Everything else unchanged:** non-predefined work offsets keep their existing behaviour.

Builds clean off `master` (CNC Controls and the full ioSender solution). One-file change to `OffsetView`.

#27 Dialogs — center owner-less modal dialogs

File	+	−
`CNC Controls/CNC Controls/MPGPending.xaml`	1	0
`CNC Controls/CNC Controls/AppConfig.cs`	1	1
`CNC Controls Probing/CNC Controls Probing/ProbeVerify.xaml`	1	0
`CNC Controls Probing/CNC Controls Probing/ProbingViewModel.cs`	1	1

Two modal dialogs — **MPGPending** (the "MPG mode active" prompt shown during the connect handshake) and **ProbeVerify** (probe-connection check) — set neither `Owner` nor `WindowStartupLocation`, so they opened as independent OS-placed top-level windows with their own taskbar button instead of centering over ioSender.

- **Owner + CenterOwner:** each now sets `Owner` to the main window at the call site and `WindowStartupLocation=CenterOwner` + `ShowInTaskbar=False` in XAML, so they center over the client area and stay attached to the app — matching every other dialog in the sender.

Builds clean off `master`. The other modal dialogs already set an owner; the owned-but-not-centered ones (g-code transform / importer dialogs) were left as-is for now.

#28 Machine Setup restructure + Tools tab + guided setup gate

File	+	−
CNC Controls/ToolsView.xaml	33	0
CNC Controls/ToolsView.xaml.cs	135	0
CNC Controls/GrblConfigView.xaml.cs	65	20
CNC Controls/IGrblConfigTab.cs	4	1

Reorganises the top-level UI into a clearer commissioning flow.

- **Settings → Grbl + App sub-tabs:** the controller ($) settings and the sender's own preferences are split into two sub-tabs instead of one long page.
- **New Machine Setup top-level tab:** an overview plus six colour-coded step-tabs, with a startup *gate* that leads a new user to the first incomplete step (step 6 queries the controller filesystem via `AtcMacros.GetStatus` for macro status). Reworks the Machine Setup Wizard (#21) into the step-tab form.
- **New Tools container:** a single home (`ToolsView`) for the commissioning/tuning tools — Stepper calibration & scratch, Surface Spoilboard (#29), Squareness (#31), Trinamic and PID — via the `IGrblConfigTab` contract; top-level Trinamic/PID dropped.
- **Inline grids:** a probe-definition grid (step 5) and a macro-status grid (step 6) surface state without leaving the wizard.

In the integration build. The bulk of the wizard body lives in #21; this entry covers the hosting/restructure (Tools container + sub-tabs + gate). Localization CSVs for the new/renamed tabs are still pending.

#29 Surface Spoilboard — machine-referenced facing tool

File	+	−
CNC Controls/SurfaceSpoilboardWizard.xaml	84	0
CNC Controls/SurfaceSpoilboardWizard.xaml.cs	589	0

Flattens the spoilboard with a boustrophedon raster, generated to the homed machine envelope.

- **Machine-referenced:** the operator touches off *Z only*; XY is referenced to the homed envelope (less an edge margin), so the raster can never overrun a soft limit no matter where Z was zeroed.
- **High-point touch-off:** jog anywhere to the board's highest point and set Z0 there; the cut still references XY to the machine corner automatically.
- **Reuse Z0 + Finish pass + abbreviated perimeter Dry run:** re-cut at the same Z0 (with a cutter/dust-boot prompt), reserve a light final pass, or trace just the perimeter to check extents.
- **Outline = leveling check:** pauses at each corner on the Z0 plane for a dial-gauge reading.
- Runs through the flow-controlled job streamer (Feed Hold / Stop stay live).

In the integration build; hardware-tested.

#30 Load Stock — corner-probe stock measurement

File	+	−
ioSender XL/LoadStockView.xaml	102	0
ioSender XL/LoadStockView.xaml.cs	721	0
ioSender XL/MachineControlPanel.xaml(.cs)	63	0
ioSender XL/MachineControlWindow.xaml(.cs)	96	0
macros/pcorner.macro	143	0

Measures real stock by probing its corners, then sets the work origin and reports geometry.

- **Corner probe via `pcorner.macro`:** an inline grblHAL NGC program calls the controller-side corner-probe macro per corner; probe feeds come from the selected 3D probe definition (no hardcoded feeds).
- **Outputs:** XY footprint, per-corner Z (flatness), squareness (skew + diagonal Δ), and stock thickness (top − spoilboard reference). A results popup draws the probed quad with corner angles.
- **Copy size:** one click puts `X Y Z` on the clipboard to paste into the Fusion `ioSenderBatchPost` add-in (#6) — round-trips the measured stock into the simulator `(STOCK)` line.
- **Floating run-control panel** (`MachineControlWindow`) so the probe run can be driven without leaving the tab.

In the integration build; hardware-tested (427×427 mm stock).

#31 Squareness (Auto Square) — ganged-gantry offset tool

File	+	−
CNC Controls/AutoSquareWizard.xaml	97	0
CNC Controls/AutoSquareWizard.xaml.cs	589	0

Facilitates Phil Barrett's auto-square *offset* method for a ganged, auto-squared gantry.

- **Drill an L:** peck-drills a reference L (corner + far-X + far-Y holes, at the framing-square arm lengths, centred on the bed with X/Y nudge to dodge dog/insert holes).
- **Measure & apply:** register a framing square on the pins, enter the gap; the tool converts it (gap ÷ lever × rail span) to a squaring-offset change, writes `$170–$172` and re-homes.
- **Ganged-axis selector:** grblHAL reports the offset for all of `$170–$172` but only the ganged axis accepts a write — the selector targets the right one.
- **Measure-only gauge:** on a build without the offset setting it still drills + measures to show how far out of square the gantry is (correct mechanically). Dry run + Reuse Z0.

In the integration build. Note: the squaring offset is fine-trim only — a grossly racked gantry must be corrected mechanically first.

#32 Program view as a reusable object + background loading

File	+	−
CNC Controls/ProgramView.xaml(.cs)	129	0
CNC Core/GCodeJob.cs	108	49
CNC Controls/GCode.cs	99	33
ioSender XL/MainWindow.xaml(.cs)	79	44
ioSender XL/LoadStockView.xaml(.cs)	68	16
CNC Controls/GCodeListControl.xaml(.cs)	29	3
CNC Controls/{AutoSquare,SurfaceSpoilboard,StepperCalibration}Wizard.xaml.cs	57	9
CNC Controls/JobControl.xaml.cs	13	7
CNC Controls/MacroProcessor.cs	0	14
macros/pcorner.macro	13	8
docs/ (2 design/session notes)	224	0
CNC Core/GrblViewModel.cs, CNC Core.csproj	11	1

“A program is a program”, taken further: the single shared program overlay becomes a standalone, streamer-connectable `ProgramView` object, and the loader stops blocking the UI.

- **Reusable program view:** Load File, Load Folder and each wizard create their own `ProgramView` and `Connect()` it to a push/pop stack — the overlay hosts whichever is active, and the loaded job is no longer a special “null” case. One Cycle Start runs the active view.
- **Continuous block numbering:** an explicit `N` word is shown in the Block column and hidden from the G-code column (Data stays intact for streaming); unnumbered lines (O-word calls, comments) continue previous + 1.
- **Background loading:** Load File / Load Folder parse on a worker thread — the per-line parse and the end-of-load bounding-box emulation run off the UI thread, blocks flush to the bound list in batches (rows appear as files are read), and a program-view-scoped wait cursor replaces the global one. A 322k-line folder combine no longer freezes the app.
- **Also shipped here:** Load Stock “set tool-length reference” option (Start Job parity) and the absolute-seek `pcorner.macro` (workaround for the G53-then-G91-on-inverted-axis firmware bug, filed upstream).

In the integration build; hardware-tested (Load Stock, single-file and folder loads, background load).

#33 Explain G-code on hover (novice tooltips)

File	+	−
CNC Core/GCodeExplainer.cs	634	19
CNC Controls/GCodeListControl.xaml(.cs)	170	20
CNC Core/CNC Core.csproj	1	0

Makes the program view teach: hovering a line shows a plain-language explanation of what the G-code does — aimed at novices.

- **Token-based:** builds the explanation from the parser's own semantic tokens (keyed to a block by line number), so it inherits the parser's modal state — a bare `X10 Y0` continuation still reads correctly as a rapid or a cut.
- **Line or selection:** hovering one line explains it ("Cutting move to X 125.4, Y 0; feed 800 mm/min"); hovering a line within a multi-row selection gives a line-by-line breakdown.
- **Full coverage:** a complete G/M-code dictionary plus a raw-line lexer fallback explain bare modal codes (e.g. `G54`) and views without parsed tokens. Computed lazily per row; skipped (thread-safe) while a background load is still parsing.
- **Reads real programs:** a bracket/parameter-aware lexer explains expression operands (`G53 G0 X[#5181]`), decodes the well-known `#5xxx` parameters (G28/G30 positions, probe result, WCS offsets), names O-word CALLs by the `/<name>.macro` file they run + their arguments, and reads G10 (WCS origin/rotation), `G65 P5` (probe-input select) and `#`-assignments plainly.
- **Click to copy:** the hover balloon is an interactive popup — click it to copy the explanation to the clipboard (a plain tooltip can't be moused into).
- **Localizable:** English for now, but every phrase is produced in one place so it can move to LibStrings later.

In the integration build; hardware-verified.

#34 Probe definition diagrams

File	+	−
CNC Controls/ProbeDefinitionEditDialog.xaml	136	22
CNC Controls/ProbeDefinitionEditDialog.xaml.cs	6	0

The probe definition editor now shows a labelled schematic beside the fields, so a novice can see what each measurement means.

- **Per-type drawing:** switched with the Type dropdown — 3D touch probe, touch plate, tool setter and edge finder each draw their shape and dimension the key measurements the user enters (tip / ball / bit diameter, plate thickness, lip width, setter height, spin speed).
- **Touch-plate lip:** drawn as a down-protruding hook whose inner face registers against the stock side (aluminium).
- **Pure vector** (Canvas + Viewbox), toggled with the same show/hide as the fields — no image assets, theme-friendly.

In the integration build.

#35 Jog distance +/− controller actions

File	+	−
CNC Core/ControllerMapper.cs	11	1
CNC Core/GrblViewModel.cs	2	0
CNC Controls/JogBaseControl.xaml.cs	1	0
CNC Controls/KeyMapEditor.xaml.cs	3	1

Two new assignable controller actions that step the on-screen jog *distance* preset (the 2×4 grid), distinct from the existing *Jog speed +/−* which step the feed preset.

- **Jog distance +/−** appear in the Controller-tab action list; assign them to any button (no default binding).
- Adds `GrblViewModel.CycleJogDistance`, wired by the jog panel to the distance index (clamped to the four presets).

In the integration build.

#36 Probe input follows the selected probe type

File	+	−
`CNC Core/Grbl.cs`	4	0
`ioSender XL/LoadStockView.xaml.cs`	6	0
`ioSender XL/HeightMapView.xaml.cs`	6	0

The program-generating tools now select the controller probe input from the chosen probe's type — the same rule the Probing page already uses — so a stale selection can't send a 3D-probe descent to the wrong input and drive into the work.

- **Rule:** Load Stock and Height Map emit `G65 P5 Q<n>` for the selected probe — tool setter → `Q1`, else the main probe → `Q0`.
- **Why:** an interrupted tool-setter cycle (`tc.macro`) can leave `G65 P5 Q1` selected; a later probe would then watch the tool-setter line and stab the spoilboard.
- Gated on new `GrblInfo.HasToolSetter` (only meaningful when a tool setter exists to select between).

In the integration build; hardware-verified (Load Stock).

#37 One in-place streaming path (stayPut removed)

File	+	−
`ioSender XL/MainWindow.xaml.cs`	49	66
`CNC Controls/MacroProcessor.cs`	12	43
`ioSender XL/LoadStockView.xaml.cs`	1	1

With each program view self-contained, the old `stayPut` flag and the "stream-in-place vs load-into-the-job" fork were a relic of the single shared job view. Collapsed to one path.

- **One streaming path:** every streamed macro/wizard run streams a transient with full flow control (Feed Hold/Stop stay live) and **never touches the loaded job** (`GCode.File`) or the Job tab. Before, wizards other than Load Stock replaced your loaded program when they ran.
- **Own view per run:** a tool that owns a program view (a wizard) marks its own; a plain streamed macro gets a dedicated run view (titled with the macro name) that auto-dismisses ~20 s after it finishes.
- **Multi-burst safe:** a run flushes several bursts (a park move, then each `O< … > CALL`); the streamer source reverts per burst, guarded so a finishing burst can't clobber the next, and the view stays connected across all of them.
- Deletes the `stayPut` param, the `RunStreamedJob` job-clobber path + its wiring, and the orphaned tab-restore helper — a net code removal.

In the integration build; hardware-verified via a clean Load Stock run (multi-burst / directives / O-words). Not yet exercised on hardware: the plain-macro run view (only a large *streaming* flyout macro reaches it).

#38 Match the bundled simulator to the connected controller

File	+	−
`CNC Controls/SimulatorManager.cs`	309	0
`ioSender XL/JobView.xaml.cs`	46	0

Keeps the bundled grblHAL simulator in step with whatever controller you connect to, so a probe / rotation / multi-axis feature behaves on the sim exactly as on the real firmware.

- **Options → signature:** on connect, the controller's behaviour-affecting build options (axis count, probe, WCS rotation) are read from `GrblInfo` into a set of compile symbols and a short 12-hex signature.
- **Cache ladder:** already-active build → locally-cached `grblHAL_sim-<sig>.exe` → a `sim-<sig>` release on the fork (public download, no token) → otherwise dispatch the parameterized `build-matched-sim` CI and pick up the result. The set of `sim-<sig>` releases acts as a shared remote build cache.
- **Token only to dispatch:** a GitHub token (`GH_TOKEN`, `workflow` scope) is needed only to trigger a build — every download is public. Runs on a background thread off `InitSystem`, never blocks connect, is skipped when talking to our own simulator, and won't re-dispatch an in-flight signature.

In the integration build; the Simulator-fork CI (`build-matched-sim.yml`) validated green end-to-end (dispatch → per-signature release → public download). The real on-connect round-trip is pending a hardware spot-check.

#39 Re-establish modal state on a mid-program start

File	+	−
`CNC Controls/StreamPump.cs`	22	0

“Start from this toolpath” queues a `G90 G94 / G17 / G21` prolog to re-establish modal state (the folder combiner strips each per-op file’s header, where `G21` lived). The legacy streamer drained that queue, but the `StreamPump` streams `source.Data` directly and never touched `source.Commands` — so the prolog was silently dropped on a mid-program start.

- **Symptom:** a mid-program start inherited whatever units the controller was left in; in `G20` the toolpath’s literal-mm coordinates read as inches → targets off the table.
- **Fix:** the pump now sends the queued `source.Commands` lines first, ahead of the first block, and swallows their acks so they don’t advance job-line accounting. Full runs from the top were unaffected (there the prolog is prepended as real blocks).

In the integration build. A latent correctness fix for any pump-based run-from-block.

#40 Load Stock “apply work rotation” defaults off

File	+	−
`ioSender XL/LoadStockView.xaml.cs`	6	1

Setting a WCS rotation via `G10 L2 R` armed a grblHAL rotation-transform bug (a rotated `G54` move with both X and Y double-applied the second axis’s offset) that soft-limited the *next* program’s first cut rapid. Reproduced on hardware: a −0.08° measured skew crashed the following job; clearing the rotation let it run.

- Default the skew→WCS rotation **off** so Load Stock never silently arms it; the checkbox stays for opt-in.
- The real fix is in the firmware rotation transform (`gcode.c`: `bit(idx)→bit(idx_0)`), now fixed and flashed; once hardware-verified the default can go back on.

In the integration build. A guard against the firmware rotation bug until the firmware fix is hardware-verified.

#41 Live 3D toolpath + material-removal carve view

File	+	−
CNC GCodeViewer/CarveView.xaml.cs	850	0
CNC GCodeViewer/Renderer.xaml.cs	190	54
CNC GCodeViewer/CarveView.xaml	36	0
CNC GCodeViewer/ConfigControl.xaml.cs	13	1
CNC GCodeViewer/RenderControl.xaml	11	7
CNC GCodeViewer/CNC GCodeViewer.csproj	7	0

A live 3D **carve view** in the toolpath viewer — not just lines on screen, but the stock being *machined away* as the program runs, so you can see the finished shape before you cut it.

- **Machine scene, always on:** the work envelope, axes and a live tool cone are shown whenever the machine is homed, even with no program loaded — a persistent sense of where the tool is in the work area.
- **Program-sized stock:** the stock block is sized to the program (its real footprint read from a `(STOCK X Y)` comment), drawn solid with the correct top, in work coordinates aligned to the toolpath.
- **Material-removal carve:** a dexel / height-field carve removes material as the tool moves, using the *real cutter bottom profile* (flat / ball / V-bit) on a fine grid, so the carved surface matches the actual tool. Toolpath lines hide during playback to reveal the cut and restore on stop.
- **Play a preview:** press Play for a machine-free preview that animates the program and builds the carve; Reset view restores the default orientation, and a pale stock theme keeps the cut visible.
- **Real machine vs. simulator:** streaming to a *real* machine **disables** the carve — the view freezes so it never competes with motion-critical streaming. To watch the carve run in step with execution, connect to the **simulator** and Cycle Start there.
- **Performance:** in-place dirty-cell mesh updates (no per-frame realloc) and the scene build deferred off the layout pass.

In the integration build. Built up over several phases (envelope / stock / tool cone → toolpath + Play → dexel carve); the validate-controller 3D visualization (#16) uses the same scene.

#42 Settings redesign — flat tab strip + searchable Grbl settings

File	+	−
CNC Controls/GrblConfigView.xaml.cs	359	56
CNC Controls/GrblConfigControl.xaml.cs	156	26
CNC Controls/KeyMapEditor.xaml.cs	66	40
CNC Controls/GrblConfigView.xaml	41	8
CNC Controls/BasicConfigControl.xaml	27	14
CNC Controls/GrblConfigControl.xaml	27	3
CNC Controls/MainPageEditor.xaml.cs	19	3
CNC Controls/MacroManagerDialog.xaml.cs	14	9
CNC Core/Grbl.cs	10	0
CNC Controls/SettingsPanelRegistry.cs	8	0
CNC Controls/AppConfig.cs	5	0
CNC Controls/AppConfigView.xaml(.cs) *(retired)*	0	373
Camera / Probing / Lathe / Jog config panels (widen + wrap)	19	10
Locale/*/csv (7 locales)	184	0

The Settings area was two tabs — **Grbl** and an **App Settings** page that was a wall of GroupBoxes plus a row of buttons that each opened a modal dialog. It's now a single **flat tab strip**: *Grbl · App · Jogging · G Code · Keyboard & Controller · Macros · Main Page*.

- **Free-text Grbl search:** a leading `$` means "setting id" — `$53` jumps to it, `$5?` expands all of `$50`–`$59`; anything else (a word like `Jog` or a bare number like `10`) matches every setting whose *name or value* contains it. Matching groups auto-expand and matches are highlighted; an **"n of m"** readout plus up/down arrows (and **F3**/**F4**, Enter) step through them.
- **Grbl-settings autosave:** an opt-in toggle (default **off** — these are `$` writes to hardware) with a companion "prompt before auto-saving"; when on, the manual Save button on the Grbl tab hides, mirroring the App-settings autosave.
- **Editors folded in as tabs:** the Key Bindings, Macros and Main Page editors were modal `Windows` launched by buttons — they're now inline tabs (converted to `UserControls`) that *save on leave* instead of OK/Cancel. The Key Bindings tab only pauses controller dispatch while it is actually shown.
- **App/Jogging/G Code:** the config panels are bucketed by type into the three tabs; App & Jogging columns are 25% wider, Camera + Probing share one column, and panels stretch to fill their column so long labels wrap instead of clipping.
- **Under the hood:** `AppConfigView` is retired — its Save/Restart footer, restart-hooking and auto-save-on-leave moved into the settings host. New strings localized across all seven locales.

In the integration build. Builds clean (Release, EXIT 0); on-hardware review of the high-DPI arrow buttons and panel wrapping pending.

#43 Per-user config folder + standalone config folded into App.config

File	+	−
CNC Controls/CNC Controls/AppConfig.cs	258	1
CNC Controls/CNC Controls/ConfigPanel.cs	97	0
CNC Controls/CNC Controls/LoadStockConfig.cs	37	0
ioSender XL/ioSender XL/Default-App.config	64	0
CNC Core/CNC Core/ControllerMapper.cs	82	27
CNC Core/CNC Core/KeypressHandler.cs	102	63
CNC Controls/CNC Controls/ProbeDefinition.cs	32	4
CNC Controls/CNC Controls/SettingsPanelRegistry.cs	43	0
Surface/AutoSquare/StepperScratch Wizard .xaml(.cs) — ConfigPanel conversion (3 wizards)	121	187
CNC Controls/CNC Controls/ConsoleControl.xaml(.cs)	162	6
CNC Controls/CNC Controls/GrblConfigControl + GrblConfigView .xaml(.cs)	143	63
Basic/StripGCode/Jog/JogUi ConfigControl + KeyMapEditor + JobControl	126	16
CNC Core/CNC Core/Grbl.cs, GrblViewModel.cs	2	2

App.config now lives per-user in `%AppData%\ioSender`: on first run the existing app-folder config (plus key maps, probe defs, macros and settings backups) is moved there, and the app-folder copy ships read-only as the reset-to-default template. The standalone `.xml` files each feature used to write next to it are folded in as `App.config` sections.

- **Section fold-in:** ProbeDefinitions, the game-controller map, key mappings, and the Surface Spoilboard / Auto Square / Stepper calibration / Load Stock parameter blobs become sections; a one-time importer reads any legacy file then deletes it. A new `ConfigPanel<T>` base gives the wizards save/restore of their section from a single DTO.
- **Curated defaults:** a shipped `Default-App.config` is the factory template (auto-saves default off) and the source for “Reset to default”.
- **Settings/console polish:** the console gains an output search bar + a font-size control; the Settings tabs share one Save/Restart/Reset footer (Grbl sub-row for reload/backup/restore/copy-to-sim); panels opt into “Reset to default”; *ResetAndUnlock* is a bindable keyboard action; a key binding reads as “modified” only when it differs from its value at startup.

In the integration build. Builds clean (Release, EXIT 0); first-run migration is best-effort and idempotent. On-hardware migration spot-check pending.

#44 Start Job workflow + Verify skew, flyout & offset fixes

File	+	−
`ioSender XL/ioSender XL/LoadStockView.xaml(.cs)`	229	127
`ioSender XL/ioSender XL/MainWindow.xaml(.cs)`	88	12
`ioSender XL/ioSender XL/JobView.xaml.cs`	3	3
`CNC Controls/CNC Controls/AtcMacros.cs`	6	19
`CNC Controls/CNC Controls/OffsetFlyout.xaml.cs`	7	3
`CNC Controls/CNC Controls/OffsetView.xaml.cs`	26	9
`CNC Controls/CNC Controls/LayoutModel.cs`	2	1
`CNC Controls/CNC Controls/LibStrings.xaml + Locale/*/csv (7 locales)`	8	8

Load Stock is reworked into **Start Job** — the first tab, the guided front end for setting a run up before the Job tab cuts it.

- **Workflow:** the operational tabs are disabled until a controller connects; startup lands on the Job tab so its handshake reads `$I`. Start Job auto-selects the configured 3D probe (no picker), zeroes at the front-left corner, and writes `start_job.macro` so the seeded macro/F-key runs the same sequence.
- **Verify skew:** a new pass re-touches each corner in the rotated work frame — the back corners twice (ideal rectangle, then the probed corner) so the gap reads out the stock's out-of-square amount — then parks at G30. The measured-skew rotation sign is corrected (grblHAL `G10 L2 R` aligns with `-atan2(dy,dx)`).
- **Flyout & offset fixes:** pinned sidebar flyouts stay open across tab switches; a flyout *Set* writes the offset directly (it was silently dropped by the streamer-state gate); the Offsets tab *Set All* skips the rapid-to-target confirmation when the machine is already there.

In the integration build. Builds clean (Release, EXIT 0). Start Job macro run + Verify skew validated on hardware; a separate 3D-view stock-orientation issue is tracked.

#45 Run-bar jog preset spinners + startup splash on top

File	+	−
CNC Controls/CNC Controls/JogPresetSelector.xaml(.cs)	106	0
ioSender XL/ioSender XL/MainWindow.xaml	11	2
ioSender XL/ioSender XL/SplashWindow.xaml	1	1
CNC Controls/CNC Controls/MDIControl.xaml(.cs)	14	6

Small quality-of-life fixes to the always-visible run bar.

- **Jog preset spinners:** a new compact selector fills the space on the *State / Check* line — *Dist [▼ value ▲] Jog [▼ value ▲] Feed*. Each field is read-only; the down/up arrows step through the four jog distance/feed presets and **wrap around** at the ends. It binds the shared `JogData` singleton, so the selection stays in sync with the jog pad, keyboard and controller (the preset values are still edited on Settings:App / the Jog tab). Right-aligned to the Program button's edge above.
- **Startup splash stays on top:** the progress splash was `Topmost="False"` and got buried by the main window's black frame almost immediately; it is now `Topmost="True"`, so the whole init sequence (connecting → reading settings → validating setup) is actually visible during the ~10–15 s startup.
- **MDI Send button re-aligned:** the MDI line was rebuilt as a single-row grid (command box + *Send*) that share one row height, and the run bar's *Bare* mode no longer pins Send to its own fixed height — so Send stays vertically centred against the editable command box (it had drifted after the high-DPI pass dropped its fixed height).

In the integration build. Builds clean (Release, EXIT 0). Splash confirmed on hardware; spinner layout and Send-button alignment confirmed in the running app.

#46 High-DPI audit — text no longer clips at 125–150% scaling

File	+	−
CNC Controls/CNC Controls/*.xaml (38 files)	114	115
CNC Controls Probing/CNC Controls Probing/*.xaml (10 files)	35	35
ioSender XL/ioSender XL/*.xaml (4 files)	18	18
CNC Controls Camera/CNC Controls Camera/*.xaml (2 files)	8	8
CNC Controls Lathe/CNC Controls Lathe/*.xaml (4 files)	7	7
CNC Converters/JobParametersDialog.xaml	6	6
CNC GCodeViewer/CNC GCodeViewer/*.xaml (3 files)	4	4
CNC Controls Dragknife/DragKnifeDialog.xaml	4	4

A reviewed, case-by-case scrub of hard-coded pixel `Width=`/`Height=` across the whole UI, so controls size to their content and no longer clip text when Windows text scaling is set to 125–150%. Roughly 250 attributes across 63 XAML files were classified one by one (not a blind find/replace):

- **Drop fixed `Height`** on text-bearing controls (buttons, labels, text boxes, combos, radio buttons, group boxes, and the rows that hold them) so they auto-size to the scaled font instead of cropping glyphs.
- **Convert fixed `Width` to `MinWidth`** (and floor heights to `MinHeight`) where the value was only a minimum — text buttons, combos, fields and container panels can now grow.
- **Convert fixed-pixel grid rows to `Height="Auto"`** where the row holds text.
- **Deliberately kept** genuine fixed geometry that must *not* scale with text: icons and images, LED/signal indicators, probe/lathe schematic diagrams, 3D render surfaces, colour swatches, icon-only glyph buttons, data-grid column widths, colon-aligned label columns, load-bearing scroll viewports and fixed dialog chrome.

In the integration build. Builds clean (Release, EXIT 0). High-traffic controls confirmed on hardware (“no more clipped text”). Known follow-ups: the five absolute-`Margin` lathe wizards (ThreadingWizard, ProfileDialog, Turning/Parting/Facing) need a real re-layout rather than an attribute scrub, and are intentionally left for a later pass.

#47 Lathe mode enable/disable checkbox

File	+	−
CNC Controls/CNC Controls/AppConfig.cs	41	0
CNC Controls/CNC Controls/BasicConfigControl.xaml.cs	14	0
CNC Controls/CNC Controls/BasicConfigControl.xaml	1	0

A *Lathe mode* checkbox on the App › Main settings panel, so lathe mode can be turned on or off without hand-editing the config. Previously the only lathe-enable control lived on a settings panel that a lathe view had to load first — and that view/tab was itself gated on lathe already being enabled, a catch-22 that made it unreachable when off.

- **Simple on/off facade** — `LatheConfig.LatheEnabled` maps to the existing `XMode` (Radius when on, Disabled when off) and persists through the same `Lathe` config section.
- **Shows/hides the Lathe wizards tab** — a new `AppConfig.SetTabPresent` adds or removes the tab in both the persisted layout tree and the legacy flat tab list, inserted after the SD-card tab.
- **Restart to apply** — the tab is only built at startup from the controller's reported lathe mode, so toggling raises `RestartRequired` (see #48).

In the integration build. Builds clean (Release, EXIT 0). Restores an implementation that had been prototyped then reverted; hardware-confirmed working.

#48 Restart button pulse + restart-on-leave prompt

CNC Controls/CNC Controls/GrblConfigView.xaml	43	1
CNC Controls/CNC Controls/GrblConfigView.xaml.cs	15	0

Makes a pending restart impossible to miss. Some settings (buffer/reset delay, 3D-view enable, main-page layout, and now lathe mode) only take effect at startup; when one changes, the shared Settings footer reveals a *Restart* button. That button now behaves like an alert:

- **Slow red pulse** — while enabled the Restart button animates between bright and dark red (custom template so the theme chrome doesn't paint over it), driven by its `IsEnabled` state so it keeps pulsing across inner-tab switches until the app is restarted.
- **Restart-on-leave prompt** — leaving the Settings area with a restart still pending asks "Restart ioSender now?" (Yes relaunches). Switching between the inner Settings tabs does not prompt.
- **Shared restart path** — the relaunch logic is factored into one `DoRestart()` used by both the button and the prompt.

In the integration build. Builds clean (Release, EXIT 0). Applies to every restart-only setting, since all route through the one shared Restart button. Hardware-confirmed ("works great").

#49 Lathe wizards rebuilt to the app layout idiom (high-DPI)

CNC Controls Lathe/CNC Controls Lathe/ThreadingWizard.xaml	275	235
CNC Controls Lathe/CNC Controls Lathe/TurningWizard.xaml	50	42
CNC Controls Lathe/CNC Controls Lathe/FacingWizard.xaml	49	41
CNC Controls Lathe/CNC Controls Lathe/PartingWizard.xaml	47	39
CNC Controls Lathe/CNC Controls Lathe/ProfileDialog.xaml	40	30

The five lathe wizards were the last upstream code still laid out on an absolute-`Margin` canvas — fixed pixel `Width`/`Height` and hard-coded positions — so their text clipped and controls overlapped once Windows text scaling was set to 125–150% (they were deliberately skipped in the #46 attribute scrub because they needed a real re-layout, not a mechanical edit). Each wizard body was rebuilt with the app's normal patterns so it flows and scales:

- **Standard idiom** — vertical `StackPanel`/`Grid` with `Auto` sizing, `NumericField` + `ColonAt` label alignment, `MinWidth`/`MinHeight` floors, a `ScrollViewer` for overflow, and grouped `GroupBox` sections.
- **Threading** — a two-column responsive grid (selection / Dimensions / Tool on the left; Thread values / Options / Cut on the right), with the linuxCNC and Mach3 option boxes sharing one slot to preserve the code-behind's draw-over behaviour, and the Tool schematics + shape radios re-hosted without absolute margins.
- **Turning / Parting / Facing** — a single scrolling input column with Calculate and the generated g-code pinned at the bottom.
- **Profile dialog** — the fixed-size modal became a `SizeToContent` window; the CSS control and fixed-RPM field share one slot, and it follows the app theme instead of a hard-coded grey.

In the integration build. Builds clean (Release, EXIT 0). Every `x:Name`, binding, converter and event handler the code-behinds rely on was preserved — no code-behind changes. The rebuilt layouts are functional and scale correctly; some aesthetic cleanup (spacing/grouping refinement) is still wanted at some point, ideally by someone who runs a lathe.

#50 Tab strips fill their width (StretchTabControl)

CNC Controls/CNC Controls/StretchTabControl.cs	119	0
ioSender XL/ioSender XL/MainWindow.xaml	6	4
CNC Controls Lathe/CNC Controls Lathe/LatheWizardsView.xaml	4	3
CNC Controls/CNC Controls/GrblConfigView.xaml	2	2
CNC Controls/CNC Controls/MachineSetupWizard.xaml	2	2
CNC Controls Probing/CNC Controls Probing/ProbingView.xaml	2	2
CNC Controls/CNC Controls/ToolsView.xaml	1	1
CNC Controls/CNC Controls/CNC Controls.csproj	1	0

The tab headers used to sit bunched at the left of each strip with dead space to the right, so tabs ran together and were hard to pick out. A new `StretchTabControl : TabControl` spreads them to fill the strip: each tab keeps its natural width (plus one space character each side) scaled up proportionally, so a wider label gets a wider tab and the row fills the width.

- **Assigns each header a width** — the default `TabControl` template hosts its headers in a fixed `TabPanel` (`IsItemsHost`), so a replacement `ItemsPanel` is ignored; setting each `TabItem.Width` is honoured instead.
- **Recomputed off-layout** — widths are recalculated only on the discrete tab add/remove event and, on resize, from a deferred (`DispatcherPriority.Background`) callback that runs *after* the layout pass. Doing it synchronously from inside layout (`SizeChanged`/`LayoutUpdated`) re-enters the layout system and mis-measures content; the deferred approach sets the widths and lets one ordinary follow-up pass consume them, with no feedback loop. The trade is that during a drag-resize the tabs show old widths for a frame and snap on release.
- **Applied to** the main tab bar and the Settings, Tools, Lathe, Probing and Machine Setup sub-tab strips.

In the integration build. Builds clean (Release, EXIT 0). Confirmed on the running app: strips fill, first tab flush, resize re-spreads cleanly, and tab content still fills/left-aligns (the label-centering was pulled back out after it leaked into content alignment on content-sized views).

#51 SD Card first-line cleanup (overlap + free-space banner)

`CNC Controls/CNC Controls/SDCardView.xaml`	6	2
`CNC Controls/CNC Controls/GrblFilesystems.cs`	3	2

Two small fixes to the SD Card tab's top line.

- **No more overlap** — the "List CNC files only" checkbox and the filesystem-status text were absolutely positioned at fixed x offsets, so the status ran under the checkbox label at larger fonts. They now flow in a horizontal `StackPanel`.
- **Banner hides when uninformative** — `FreeSpaceSummary` only lists a filesystem that actually reports free space; the bare "*name*: mounted" fallback carried no information (the file list already shows the filesystem is present) and just added clutter, so the whole banner disappears when the controller reports no sizes.

In the integration build. Builds clean (Release, EXIT 0). Confirmed on the running app.

#52 Per-tool guidance text on the Tools tabs

File	+	−
CNC Controls/CNC Controls/ToolView.xaml	3	0
CNC Controls/CNC Controls/SurfaceSpoilboardWizard.xaml	11	5
CNC Controls/CNC Controls/StepperCalibrationScratchWizard.xaml	7	1
CNC Controls/CNC Controls/StepperCalibrationWizard.xaml	8	2
CNC Controls/CNC Controls/AutoSquareWizard.xaml	4	4
CNC Controls/CNC Controls/TrinamicView.xaml	8	2
CNC Controls/CNC Controls/PIDLogView.xaml	5	0

Most of the Tools tabs showed a form with no explanation of what the tool did or how to run it. Each now carries a short “About…” heading + wrapped explainer in the tab’s right-hand pane (matching Auto Square and Stepper Calibration, which already had step text).

- **Covered** — Tool table, Surface spoilboard, Stepper calibration (scratch), Trinamic tuner, PID tuner.
- **Layout tidy** — the Stepper-calibration collision warning is contained to the right column (instructions box fills the height); the Surface-spoilboard result sits in its own light panel distinct from the description; Squareness puts the description above the diagram.
- **Localizable** — every string carries an `x:Uid` so it feeds the locale CSVs; English inline is the fallback.

In the integration build. Builds clean (Release, EXIT 0). Confirmed on the running app.

#53 Headless build script + crash logging

File	+	−
`build.ps1`	105	0
`.vscode/tasks.json`	69	0
`ioSender XL/ioSender XL/App.xaml.cs`	72	7
`CLAUDE.md`	9	1

Build and run the WPF app entirely from the command line / VS Code, without opening the Visual Studio GUI. (PlatformIO can't drive a .NET Framework build — it only builds the C/C++ firmware/sim — so a plain MSBuild wrapper is the right tool.)

- **`build.ps1`** — kills a running `ioSender.exe`, builds via MSBuild (located with `vswhere`, Enterprise-path fallback), optionally launches. `-Configuration Debug|Release|Both`, `-Launch`, `-Headless`, `-NoKill`.
- **VS Code tasks** — *Build Debug + Launch* (Ctrl+Shift+B), *Build Release*, *Build Both*, all delegating to the one script.
- **Crash logging** — the global handlers (Dispatcher / AppDomain / TaskScheduler) now dump a timestamped, versioned, full-stack entry to `%AppData%\ioSender\ioSender.crash.log` and exit `0xFA11` (64017). Fixes a real bug where a background-thread exception showed a box but never exited, wedging the process. Interactive runs still show a dialog; `IOSENDER_HEADLESS=1` suppresses it so an unattended crash can't hang on an un-clicked box. VS's JIT debugger does not auto-attach.

In the integration build. Debug + Release build clean; both crash paths verified live (log written, exit 0xFA11).

#54 Rename LoadStock → Start Job (code matches the UI)

File	+	−
`ioSender XL/ioSender XL/StartJobView.xaml(.cs)` *(was LoadStockView)*	9	9
`CNC Controls Probing/…/StartJobControl.xaml(.cs)` *(was LoadStockControl)*	9	9
`CNC Controls/CNC Controls/StartJobConfig.cs` *(was LoadStockConfig)*	10	10
`CNC Controls/CNC Controls/AppConfig.cs`	6	6
`CNC Controls/CNC Controls/LayoutModel.cs`	3	3
`CNC Controls/CNC Controls/{ICNCView,LibStrings,StreamPump}`	4	4
`ioSender XL/…/{MainWindow,JobView}.xaml.cs + IProbeTab.cs`	3	3
`Locale/*/csv (7 locales, 2 files each)`	126	126
`*.csproj (3 projects)`	7	7

The feature is "Start Job" everywhere in the UI, but the code was still `LoadStock*`. Full rename — there is no back-compat to preserve, so no migration code.

- **Files & classes** — `LoadStockView` / `Control` / `Config` / `Settings` → `StartJob*`.
- **Tokens** — `ViewType.LoadStock` and `ProbingType.LoadStock` → `StartJob`; `LayoutKeys` value, the folded App.config section (`LoadStock` / `LoadStock.xml`), the `str_tabLoadStock` resource key and its locale-CSV baml paths, and the StreamPump temp-log name all follow.
- **Persisted config** — the three tokens saved in `%AppData%\ioSender\App.config` (tab name, layout component, section key) were patched out-of-band to match.

In the integration build. Debug + Release build clean; headless startup smoke test passes on the renamed config.

#55 Localization sweep — V2 strings into all 7 locales

File	+	−
`Locale/*/csv/CNC.Controls.WPF.resources.*.csv` *(7 locales)*	1300	0
`Locale/*/csv/CNC.Controls.Probing.resources.*.csv` *(7 locales)*	147	0
`Locale/*/csv/ioSender.resources.*.csv` *(7 locales)*	63	0
`Locale/*/csv/CNC.Controls.Viewer.resources.*.csv` *(7 locales)*	35	0
`tools/locadd.py`	36	1

Later V2 features were `x:Uid`'d in XAML but never had LocBaml rows added, so they fell back to English in the other six languages. The self-deriving `tools/locadd.py` was pointed at the current view set and backfilled every missing row.

- **+1545 rows across 7 locales** — 159 new keys in en-US plus the backfills for keys that had reached en-US but not the other locales, so all seven are consistent again.
- **Covers** — Start Job (renamed view + the probing control), the probing-redesign dialogs (`ProbeMotionParams` / `ProbeDefinition*`), the main-page editor, console search, the Tools tab guidance (#52), the carve view, and assorted flyouts/config controls.
- **Baseline, not machine-translation** — each new row carries the English text for a translator to replace; the app already rendered English via fallback, so nothing changes on screen. Comma values RFC4180-quoted; re-running the tool adds nothing.

In the integration build. Lathe wizards are excluded (no `CNC.Controls.Lathe` locale CSV exists — never LocBaml-localized).

#56 Online user manual + in-app context help (F1)

File	+	−
`CNC Controls/ManualHelp.cs` *(new)*	112	0
`ioSender XL/MainWindow.xaml.cs`	13	2
`CNC Controls/CNC Controls.csproj`	1	0

A new task-oriented **online user manual** — a single self-contained HTML page hosted on GitHub Pages at `iosenderv2.github.io/ioSender` — plus an in-app hook that opens it *in context*.

- **Context help** — `F1` opens the manual at the page for whatever tab is current (a `ViewType`→anchor map in `ManualHelp`, resolving a bundled copy next to the exe or the online copy). The Help menu's *Usage tips* and *A brief tour* are repointed from the upstream wiki to the V2 manual.
- **The manual** (in `docs/manual/`, published from an isolated `gh-pages` branch) — 18 topics threaded into three reading tracks (novice / intermediate / machinist), an A–Z subject index, live search, three hand-built inline SVG diagrams (MCS-vs-WCS, machine anatomy, tool-change flow), 13 screenshots, and a lazy "Watch on the machine" video hook for future walkthroughs.

In the integration build. The manual site + its authoring tools (`docs/manual/*`) are not enumerated above (self-referential docs).

#57 Fixes: Start Job tab strand + program-view overlay width

File	+	−
`ioSender XL/MainWindow.xaml.cs`	21	0
`ioSender XL/JobView.xaml.cs`	4	0
`ioSender XL/MainWindow.xaml`	1	1

Two rough edges surfaced while documenting the Start Job flow.

- **Stranded tab** — tab enablement was driven by fragile `Hold` / `IsGCLock` event pairs (Start Job's auto-probe locks G-code); a mis-paired clear could leave a connection-gated tab (e.g. Start Job) disabled with no way back. New `RefreshTabsIdle()` reconciles every tab to its correct connection + capability state whenever the controller is idle and nothing is streaming — idempotent, and it never runs during a real job.
- **Program-view overlay** — the generated-program overlay had no maximum width, so it grew to cover the whole tab and hid the Start Job inputs; capped with `MaxWidth="720"` so it stays a right-hand column.

In the integration build. Both regressions trace to the run-bar / StretchTabControl era (#45, #50).

#58 Numeric fields select their value on focus

File	+	−
CNC Controls/NumericTextBox.cs	35	0

Tabbing or clicking into any numeric field now selects the whole value, so you can just type the replacement instead of clearing the old number first.

- Applied once in the shared `NumericTextBox`, so it covers every numeric field app-wide (DRO, Settings, Probing, Start Job, jog step, the wizards…).
- Keyboard / tab focus selects immediately; a focusing click lets WPF place the caret natively then selects on mouse-up (plain click only, so a click-drag still keeps your partial selection). Once the field already has focus, later clicks fall through untouched — so editing a single digit still lands the caret where you click.

In the integration build.

#59 Connect: Network default + LAN controller discovery

`CNC Core/NetworkScanner.cs`	349	0
`CNC Controls/PortDialog.xaml.cs`	127	3
`CNC Controls/PortDialog.xaml`	32	6
`Locale/*/csv (7 locales)`	19	0

The Connect dialog now opens on the **Network** tab by default (most grblHAL controllers are networked), and a new **Scan network** button finds controllers on the LAN so you don't have to know the IP.

- **Discovery by protocol, not open ports.** An async subnet sweep (up to a /24 per interface, 64 concurrent) probes each host and confirms it's really a grbl controller by sending the real-time status request `?` and requiring a valid `<State|…>` report back — then reads `$I` to show "grblHAL <version>". Random devices with telnet open are rejected. mDNS (`grblHAL.local`) is tried first but not relied on: consumer WiFi routers routinely drop mDNS multicast between the wireless and wired segments, so the subnet scan is the real workhorse.
- **Editable host picker.** The IP field is now a combo — type a host, or pick one the scan found (the dropdown shows "host:port — grblHAL x.x", the edit box shows just the host, and OK connects to whatever is current).
- The dialog also no longer hides behind the startup splash (it docks to the top of the screen while the splash is up), and its fields no longer clip at high DPI.

In the integration build.

#60 Drag to reorder tabs (persisted across restarts)

CNC Controls/StretchTabControl.cs	178	0
CNC Controls/TabOrderConfig.cs	48	0
CNC Controls/AppConfig.cs	56	0
ioSender XL/MainWindow.xaml.cs	13	1
CNC Controls/ToolsView.xaml.cs	5	1
CNC Controls Probing/ProbingView.xaml	5	5
CNC Controls/GrblConfigView.xaml	1	1

Grab any tab **header** and drag it left or right to change tab order — on the main bar and on every sub-tab strip (Settings, Tools, Probing). A plain click still just selects the tab; the drag only starts once the pointer passes the system drag threshold, and the cursor switches to a ↔ while dragging. The new order now **survives a restart**.

- **Persisted to whichever store already governs a bar — never a second source of truth.** The Probing and Settings strips (which had no order config before) are saved by the control itself into a new `TabOrder` section of `App.config`, keyed per strip. The main bar and the Tools sub-tabs are already backed by the layout tree, so a drag there writes into `Config.Tabs` / the layout tree — the *same* store the **Edit Main Page → Tabs** editor uses, so dragging and the editor always agree (and drag takes effect immediately, no restart needed).
- **Capability-hidden tabs stay put.** Tabs hidden for the current controller (e.g. Lathe Wizards outside lathe mode, Trinamic / PID when unsupported) keep their stored position — only the on-screen tabs reorder among themselves, so a hidden tab can never be dropped from the saved layout.

In the integration build.

#61 grbl:Settings tree populates when opened after connecting

CNC Core/Grbl.cs	12	1
CNC Controls/GrblConfigControl.xaml.cs	25	3

Opening **Settings › Grbl** after connecting to a grblHAL controller could show an empty settings tree — no settings at all, and a $-search found nothing. The flat-tab Settings redesign builds the settings control eagerly at startup, and during the connect handshake the Grbl tab gets an early activation *after* the controller answers `$$` but *before* `$I` establishes that it reports setting enums. The settings load then took the plain `$$` path and populated the list group-less, skipping the setting-details/group query that builds the tree — and because the list was now non-empty, every later load skipped that query too, so the group tree was never built.

- **Build the metadata when it's finally available.** The settings load now also fetches the setting-enum details and group tree when the controller reports enums but the tree hasn't been built yet, dropping the stale group-less entries first so they're rebuilt with full name/type/group metadata (no duplicates).
- **Reach the load on entry.** The Grbl tab no longer short-circuits its activation when it was pre-activated (empty) during the disconnected startup pass, so opening the tab after connecting always (re)builds the tree.

In the integration build. Fixes a regression from the Settings flat-tab redesign (#42); supersedes an earlier partial fix that only re-bound the (still-empty) tree.

#62 Probing & Tools sub-tabs share one layout

CNC Controls Probing/EdgeFinderControl.xaml	10	8
CNC Controls Probing/EdgeFinderIntControl.xaml	10	8
CNC Controls Probing/CenterFinderControl.xaml	10	8
CNC Controls Probing/ToolLengthControl.xaml	6	4
CNC Controls Probing/ProbingView.xaml	1	1
CNC Controls/StepperCalibrationWizard.xaml	1	1

A consistency pass so the descriptive/instructions area and the results area look the same across the four Probing sub-tabs (Tool length offset, Edge finder external/internal, Center finder) and the Stepper calibration tool, matching the newer Tools tabs: **instructions on the ambient window grey, results in a white panel**, with clear separation.

- **Instructions on the ambient grey, not a near-white box.** The instruction text sits directly on the window grey (as the Tools tabs do), and each tab's output (position / message / preview) is a bordered white panel.
- **The real culprit was the tab host.** The Probing sub-tabs render inside a `StretchTabControl` that — unlike the Tools one — never set `Background="Transparent"`, so the stock TabControl template painted its content area white behind the transparent instructions. Making the Probing host transparent lets the ambient grey show through, so all four tabs finally match.

In the integration build.

#63 Check mode ($C) moved to a Cycle Start context-menu item

CNC Controls/JobControl.xaml.cs	22	0
CNC Controls/JobControl.xaml	11	1
CNC Controls/StatusControl.xaml.cs	1	11
CNC Controls/StatusControl.xaml	0	10
Locale/*/csv (7 locales)	14	7

The **Check** checkbox next to the *State* field on the status bar was misplaced and rarely used. It's gone from the status bar; the grbl check-mode toggle is now a checkable **"Check mode ($C)"** item on the **Cycle Start** button's right-click menu — a dry-run / validate toggle now lives with the button that starts a run.

- **Same semantics.** The menu item mirrors the live check-mode state, is enabled only when not running a job and not asleep, sends `$C` from Idle to enter check mode, and soft-resets to leave it — exactly as the old checkbox did.
- **Localized.** The new menu string (header + tooltip) is added to all 7 locale CSVs; the retired checkbox label rows are removed.

In the integration build.

#64 Opt-in diagnostic trace log (`-debuglog`)

`CNC Core/DebugLog.cs`	136	0
`ioSender XL/App.xaml.cs`	36	5
`CNC Controls/GrblConfigControl.xaml.cs`	6	0
`CNC Core/CNC Core.csproj`	1	0

A lightweight, always-compiled-in diagnostic logger — `CNC.Core.DebugLog`. It is **off by default** and costs a single boolean test per call site when disabled, so `DebugLog.Write(…)` statements can be left permanently at the tricky points in the code rather than being hand-added and torn down each time a layout / flow bug needs tracing. Diagnosing the earlier empty-settings-tree regression (#61) meant hand-rolling a throwaway logger; this makes that capability standing.

- **How it's turned on.** Launch with `-debuglog` (all categories) or `-debuglog=settings,comms` (an allow-list); an `IOSENDER_DEBUGLOG` env var does the same for unattended/headless runs (mirrors `IOSENDER_HEADLESS`).
- **Where it writes.** Timestamped lines (`HH:mm:ss.fff  [category]  message`) to `%AppData%\ioSender\ioSender.debug.log`, next to the crash log, with a per-run banner (version + categories) and automatic roll at 8 MB. Thread-safe append, and never throws — logging can't take the app down.
- **First call sites.** The fragile grbl:Settings activation path (`Activate` / `ConfigureView`) now traces its `HasEnums` / `IsLoaded` / setting & group counts, so a recurrence of the empty-tree symptom is visible in the log immediately.

In the integration build. Available in every configuration (not Debug-only), gated by the flag.

#65 grbl:Settings premature activation prevented at source

CNC Controls/GrblConfigView.xaml.cs	19	0

The root trigger behind the empty-settings-tree saga (#61), now closed off at the source. The Settings view's inner tab strip raises an initial `SelectionChanged` for its default tab during eager startup layout — before the view is ever shown and, on a network connect, mid-handshake. That fired a premature `Activate(true)` on the Grbl tab while `$I` hadn't yet reported enum support (`HasEnums == false`), which is exactly what loaded the settings group-less.

- **The fix.** A `_viewActive` flag, set only by the top-level `Activate(bool, ViewType)`. `tab_SelectionChanged` now ignores inner selection churn until the view is genuinely active — the top-level `Activate(true)` already enters the current tab itself, so real user tab switches still work, but the spurious startup fire is dropped.
- **Belt-and-suspenders** over the #61 `Load()` guard (which rebuilds the tree metadata whenever it's missing): even if an early activation slipped through, the tree would still build — but now it can't fire disconnected in the first place. Verified with the #64 trace log: the mid-handshake `Activate` line is gone; the first and only activation lands post-`$I` with the groups already populated. Also removes wasted startup work.

In the integration build.

#66 Startup flag `-forgetNetwork`

`CNC Controls/AppConfig.cs`	13	0

A small startup/testing aid: launch with `-forgetNetwork` to **ignore the saved connection target for that run** and open the connection dialog instead of auto-connecting the remembered host — from there you can scan the network or pick any transport (serial / network / simulator).

- **Non-destructive.** Nothing is written to `app.config` unless you then choose a target, so the remembered value survives; on the next serial connect it's re-learned from the controller's `$I`-reported IP anyway. It reuses the existing `-selectport` path to force the dialog.
- **Why.** The modal connect dialog delays the handshake, which makes it a clean, repeatable way to reproduce startup-timing bugs (such as the #65 activation race) without hand-editing the config. The connect path also gained a `[connect]` trace line (see #64) reporting the saved target and flags.

In the integration build.

#67 Demo-video capture markers (`-demomarker`)

CNC Core/DemoMarker.cs	94	0
CNC Core/GrblViewModel.cs	1	0
CNC Core/CNC Core.csproj	1	0
ioSender XL/App.xaml.cs	14	0
ioSender XL/MainWindow.xaml	32	1
ioSender XL/MainWindow.xaml.cs	76	0

Production aids for recording the manual's demo videos — opt-in via `-demomarker` (or the `IOSENDER_DEMOMARKER` env var) and invisible in normal use. A sibling of the #64 trace log; the full capture-and-composite procedure lives in `docs/demo-videos/`.

- **Marker log.** `CNC.Core.DemoMarker` writes a small CSV (`%AppData%\ioSender\ioSender.demo-markers.csv`, started fresh each launch) of timestamped events: `SESSION_START`, `RUN_START`/`RUN_END` (from the single `IsJobRunning` edge transition), and `TIMELAPSE_ON`/`TIMELAPSE_OFF`. A video compositor keys off these to *sync* the app screen-recording to the machine-camera footage and to speed up the timelapsed stretch — no hand-keyframing.
- **Timelapse toggle.** A run-bar toggle (shown only under `-demomarker`, beside the State field) marks the timelapse window live — flip it as you switch the camera to / from timelapse. Traffic-light coloured: green = off (press to start), red = running (press to stop). Fallbacks if you never flip it: auto-on a few minutes after Cycle Start, forced off at job end.
- **Near-free when off:** a single boolean test per call site — nothing is written and the toggle is collapsed on normal runs.

In the integration build. Builds clean (Debug + Release).

#68 OBS auto-record bridge + tool-change cut markers

CNC Core/ObsBridge.cs	168	0
CNC Core/GrblViewModel.cs	13	0
CNC Core/CNC Core.csproj	1	0
ioSender XL/App.xaml.cs	14	0
ioSender XL/MainWindow.xaml.cs	2	0
build.ps1	13	3

Extends the #67 demo facility so a shoot can be hands-off. When armed with `-demomarker`, ioSender connects to **OBS Studio**'s built-in **obs-websocket** server (v5 handshake, optional auth via `IOSENDER_OBSWS_PASSWORD`, defaults to `localhost:4455`) and drives recording from job state — no manual Record button. Reuses the websocket-sharp dependency already in the build; never throws and no-ops if OBS isn't reachable.

- **Auto record. Program load** → `StartRecord`; **program end** (unload) → `StopRecord`. The recording spans the whole session (setup, probing, every cut, tool changes), so it doesn't stop between phases.
- **Cut markers refined.** `RUN_START`/`RUN_END` now mark the *actual cutting*. Entering the `Tool` state (an M6 tool change) ends the current cut and leaving it starts the next — i.e. `…RUN_END <tool-change> RUN_START…` — so the compositor can treat cutting spans and tool-change gaps differently.
- **Launch pass-through.** `build.ps1` now forwards any trailing tokens to the launched exe, e.g. `.\build.ps1 -Launch -demomarker -forgetnetwork`.

In the integration build. Builds clean (Debug + Release). Camera-agnostic — drives OBS, so it works with any source (RTSP cams, screen capture). See `docs/demo-videos/`.

#69 Per-Monitor V2 DPI awareness

`ioSender XL/app.manifest`	33	0
`ioSender XL/MainWindow.xaml.cs`	40	0
`ioSender XL/App.config`	5	0
`ioSender XL/ioSender XL.csproj`	1	0

ioSender was **System-DPI-aware** (no manifest) — it used a system-DPI value cached at logon, so a resolution / scale change made in the current session (or a stale per-monitor override) could leave it rendering at the wrong scale until the next sign-out. It is now **Per-Monitor V2** aware, reading the *live* per-monitor DPI like Edge and VS Code.

- **How.** An `app.manifest` declaring `dpiAwareness = PerMonitorV2` (referenced from the csproj), plus the WPF `Switch.System.Windows.DoNotScaleForDpiChanges=false` app-config switch so the visual tree re-scales on a live DPI change.
- **Diagnostics.** A `-debuglog=dpi` trace (see #64) logs the resolved `DpiScale` / device transform / screen & window metrics at startup and on every `DpiChanged`, so a "wrong size" report can be pinned to a scale value in seconds.
- **Note.** Interop surfaces (3D carve view, camera) should be spot-checked across a live DPI change; the switch can be flipped back to keep launch-time correctness without live re-scaling if needed.

In the integration build. Builds clean (Debug + Release). Verified at 100% / 175% / 300% — the app tracks the live monitor scale.

#70 Named G28 fixtures

CNC Controls/NamedPositionConfig.cs	77	0
CNC Controls/OffsetFlyout.xaml.cs	215	3
CNC Controls/GotoBaseControl.xaml.cs	24	0
CNC Controls/OffsetFlyout.xaml	3	0
CNC Controls/AppConfig.cs	3	0
CNC Controls/CNC Controls.csproj	1	0
Locale/*/csv/CNC.Controls.WPF (7 locales)	13	0

A saved library of **named machine-coordinate positions** (“fixtures”) recalled from the **G28** sidebar flyout — for recurring jigs like a 90° sheet-goods fence origin or a machinist-vice corner. grblHAL stores only a single G28 slot, so the library lives app-side (in machine coordinates) and never clobbers it.

- **In the flyout.** An editable, alphabetically-sorted name dropdown sits left of the *Go* button (G30 / G59.3 flyouts are unchanged — those are deliberate out-of-envelope park spots, not fixtures). **Set** stores the current machine position under the box’s name; **Go** rapids to the selected fixture with the same Safe-Z lift as the G28 button (a new `GotoBaseControl.SafeGotoMachine`, app-side `G53`), falling back to the firmware G28 when no fixture is named.
- **Position-aware.** On focus the box reconciles the name to where the machine actually is — the matching fixture, else the next `Fixture-N` default — so Set saves the right thing without hunting.
- **One name per position.** Set renames / dedupes / updates in place, and duplicates left by earlier sessions self-heal on launch (keeping your custom name over an auto `Fixture-N`).

In the integration build. Builds clean (Debug + Release). Hardware-verified. Loc: `cbo_fixtures` tooltip in all 7 CNC.Controls.WPF CSVs.

#71 Smoother, faster corner probing

`macros/probe_tfl.macro`	18	11
`macros/pcorner.macro`	12	11

Two refinements to the Start Job corner-probe macros (`probe_tfl` / `pcorner`).

- **No more table shudder.** Every post-trigger move — the pull-off between the coarse and latch probes, and the retract after the latch — was a `G0`, so it ran at max rapid *acceleration*: a violent jerk on a 4–10 mm move that shook the whole gantry. They are now `G1 F1000`: identical distance and accuracy (the measurement is the slow `G38.2` latch), but a bounded peak velocity so the ramp is gentle.
- **≤ 1" stock optimization.** The spoilboard probe now rapids straight down to the stored G28 Z (the same sequence as the G28 flyout button) and seeks only a short capped distance; the stock-top probe starts just above the tallest possible 1" top instead of machine-top, cutting its seek from ~100 mm to ~32 mm. Documented as preconditions: stock ≤ ~25 mm and G28 Z set just above the spoilboard.

In the integration build. Hardware-verified ("smooth as silk"). On-controller macros — re-uploaded to littlefs by Start Job's checksum provisioning.

#72 Start Job: estimated stock size & Z-thickness check

ioSender XL/StartJobView.xaml.cs	13	0
ioSender XL/StartJobView.xaml	5	2
CNC Controls/StartJobConfig.cs	1	0
Locale/*/csv/ioSender.resources (7 locales)	28	14

The Start Job setup fields are relabelled **Est.** (they are an estimate a measure run refines), and a new **Est. thickness (Z)** field is added. Since the probe macros now assume stock ≤ 1″ (#71), a red warning appears under the field when the Z estimate exceeds 25.4 mm — live and on load. The estimate persists with the other Start Job inputs; it is advisory only.

#73 Playbooks: one-stop procedure library

`docs/playbooks/*.md (14 files)`	456	0

The repeatable development procedures — previously scattered through dev notes — are collected into a single checked-in `docs/playbooks/` folder, **one procedure per file**, each with the ordered steps plus a ready-to-run command or copy-paste fragment (substitute the parameters and go). `README.md` indexes the set.

- **Docs & release:** add a changelog entry (the four places to touch + the detail/TOC HTML skeletons), regenerate `Overview.pdf`, publish the manual site, re-import a manual screenshot, wire a video into the manual.
- **Build & ship:** the build/commit/test loop, the headless `build.ps1`, the Teensy firmware build, the end-of-session wrap-up sequence and conversation-log capture.
- **Cross-cutting:** the localization catch-up pass, running `gh` / the GitHub API on this box, and compositing a demo video.

Documentation only — no code change. Each playbook cites the repo script it drives (`build.ps1`, `tools/locadd.py`, `docs/manual/publish-pages.ps1`, …) as the authoritative implementation.

#74 Startup dialogs no longer hidden behind the splash

`ioSender XL/SplashWindow.xaml(.cs)`	14	1
`CNC Controls/PortDialog.xaml.cs`	4	16

The startup progress splash was `Topmost` and centred, so any dialog that opened during startup was mostly hidden behind it. That was previously worked around for the connect dialog alone by docking it to the top of the screen — but other startup popups (message boxes, machine-setup prompts, the "controller never reported" safety net) still landed behind the splash.

- **Fixed at the source:** the splash now docks itself near the top of the work area, leaving the centre clear for *every* dialog. The connect-dialog special-case is removed — it simply centres on screen again (still owns to the splash so it wins z-order).

#75 Status-bar messages briefly shown at double height

`ioSender XL/MainWindow.xaml.cs`	37	0

The status line at the bottom of the window is short and easy to miss. When a new message is written to it, it is now shown at **twice its normal height for 10 seconds** (or until the next message re-triggers the enlarge), then shrinks back — so an important message stands a chance of being noticed. An empty/cleared message reverts immediately, and the base size tracks the theme/DPI font scaling.

#76 Program view: 3-line run view on Cycle Start

CNC Controls/GCodeListControl.xaml.cs	90	0
CNC Controls/ProgramView.xaml(.cs)	66	2
ioSender XL/MainWindow.xaml.cs	14	0

When a wizard *Generate* pops its program view open and you press **Cycle Start**, the view now collapses to a compact **3-line run view** — previous line, current executing line (green), next line — kept centred on the executing line as the run advances, and dropped to the **bottom** of the tab content area (flush above the run bar) so a running program takes little space and stays in the line of sight.

- **Toggle:** click anywhere on the title bar to switch between the compact run view and the full program (hand cursor + hint text make it discoverable).

#77 Selected tab stands out — bold, slightly larger label

`CNC Controls/StretchTabControl.cs` 62 1

The active tab was distinguished only by its white background, easy to lose track of on the wider strips. The selected tab's **label** now also goes **bold and ~18% larger** so the current tab reads at a glance. Applies to every bar built on `StretchTabControl` (main bar + Settings / Tools / Lathe / Probing / Machine Setup) with no per-view change — see #50.

- **Header only:** the emphasis is set on the tab's header presenter, not the `TabItem` — a `TabControl` re-parents inheritance so font set on the selected `TabItem` would bold the whole page; confining it to the header leaves the page content untouched.
- **No reflow:** each tab keeps its already-assigned width, so enlarging the selected label doesn't retrigger the stretch recompute — the bolder text simply fills its fixed slot (a touch cramped when selected, by design).

#78 Run-bar jog pad restored & the button row rebalanced

`ioSender XL/MainWindow.xaml`	34	33
`CNC Controls/StatusControl.xaml`	5	3
`CNC Controls/JobControl.xaml`	3	3

The shrunk jog pad in the bottom run bar had disappeared: it lived on the leftover space right of the run buttons, and the **Program** toggle added to that row (the program-overlay work) widened it and consumed the leftover, collapsing the pad to zero width.

- **Four-column run bar:** *button row · jog pad · signals+feed · overrides*. The button row is the flexible (star) column and stretches to fill the slack up to the jog pad — its three groups spread across so the run buttons span the width instead of huddling left, with the MDI and State lines following so *Send* stays aligned with *Program*. The jog pad, signals and feed/rapids overrides are content-sized and ride at the right edge, adjacent.
- **Reclaimed the width:** run buttons are now content-sized (one space of padding each side) instead of a fixed 66 / 95 px, and the Signals + override clusters were shrunk ~25%, so the jog pad has room.

#79 Bind any tab to a key — right-click Bind-to-Key + header badges

File	+	−
CNC Controls/TabKeyBinder.cs	259	0
ioSender XL/MainWindow.xaml.cs	122	4
CNC Controls/KeyMapEditor.xaml.cs	142	2
CNC Controls/GrblConfigView.xaml.cs	46	1
CNC Controls/MachineSetupWizard.xaml.cs	35	0
CNC Controls Lathe/LatheWizardsView.xaml.cs	27	1
CNC Controls Probing/ProbingView.xaml.cs	27	1
CNC Controls/ToolsView.xaml.cs	27	2
CNC Controls/AppConfig.cs	19	0
CNC Controls/MachineSetupView.xaml.cs	7	1
CNC Controls/KeyMapEditor.xaml	2	3
CNC Controls/ComponentAvailability.cs	1	1
CNC Controls/LibStrings.xaml	1	1
CNC Controls/CNC Controls.csproj	1	0

Every main-page tab *and* its second-level sub-tabs can now be bound to a keyboard shortcut, so you jump straight to a tab from anywhere instead of hunting through the strip. Builds on the in-app Key Mappings editor (#3).

- **Right-click any tab → Bind to Key** (or *Change Key / Clear Shortcut*): a small capture modal takes one combination — no need to open the editor and find the action. The current shortcut shows as a dim **badge in the tab's upper-right corner**, live-updating.
- **Full coverage:** the 10 main tabs plus the inner tabs of *Settings* (7), *Probing* (4), *Tools* (7), *Machine Setup* (Overview + 6 steps) and *Lathe Tools* (4). Nested `Tab.<Parent>.<Sub>` ids select the parent tab then the inner one through a shared `ITabBindingHost`; bindings persist in `Config.TabShortcuts` and dispatch at the window level so they fire regardless of focus.
- **Smart no-ops:** a stale binding to a tab that has since become unavailable (capability removed, disconnected, mode disabled) does nothing at all rather than half-switching to the parent. Text-producing combos (no modifier, or Shift only) are never stolen from a focused text box.
- **Editor parity:** the Key Mappings list groups every tab by parent; tab rows highlight when bound (changed from the unbound default), the same whether set via right-click or in the editor, and re-sync on show.
- Also fixed a pre-existing key-capture bug in the editor (a stray `FocusManager.IsFocusScope` swallowed keypresses) and switched its list to line-by-line scrolling. Renamed the *Lathe Wizards* tab label to **Lathe Tools**.

#80 Capability-gated tabs own their own “unavailable” reason

File	+	−
CNC Controls/IAvailabilityGated.cs	26	0
CNC Controls/StretchTabControl.cs	23	0
CNC Controls/ComponentAvailability.cs	21	32
CNC Controls/ToolsView.xaml.cs	7	33
ioSender XL/JobView.xaml.cs	7	21
CNC Controls/AutoSquareWizard.xaml.cs	9	1
CNC Controls Probing/ProbingView.xaml.cs	8	1
CNC Controls/SDCardView.xaml.cs	6	1
CNC Controls/ToolView.xaml.cs	6	1
CNC Controls/PIDLogView.xaml.cs	5	1
CNC Controls/TrinamicView.xaml.cs	5	1
CNC Controls Lathe/LatheWizardsView.xaml.cs	5	1
CNC Controls/CNC Controls.csproj	1	0

When the connected controller lacks a capability, the tab that needs it is removed and listed — with the reason — under *Edit Main Page › Unavailable*. That reason used to be a hand-maintained switch kept *separately* from the code that actually removes the tab, so the two drifted: the list claimed SD Card needs an SD card while the real test was “any filesystem” (SD *or* LittleFS).

- **One source of truth:** each gated view now owns its own prerequisite + message through a new `IAvailabilityGated` interface (Probing, Lathe Tools, SD Card, Tool table, Trinamic, PID, Auto Square). The removal and the listing read the same property, so they can’t disagree.
- **No central list:** `StretchTabControl.PruneUnavailable()` — used by both the main tab bar and the Tools sub-bar — drops the unavailable tabs and records their reasons; the old string switch is gone.
- Fixes the SD Card mismatch (both now agree on “has a filesystem”), and Auto Square stays as a squareness gauge when the offset setting is absent, with the limitation noted rather than the tab removed.

Internal refactor; no visible behaviour change beyond the corrected SD Card reason. Release build verified.

#81 No more restart prompt hiding behind the relaunch splash

File	+	−
`CNC Controls/GrblConfigView.xaml.cs`	18	1

Applying a restart-only change (e.g. toggling lathe mode) and relaunching could leave a "Restart ioSender now?" message box stranded *behind* the new instance's splash screen — it only surfaced after the freshly launched copy went away, looking like the app prompted *after* it had already restarted.

- **Cause:** the relaunch launches the replacement (splash on top) and then shuts down, and the shutdown path re-enters the Settings "on leave" logic — which, seeing a restart still pending, popped the prompt a *second* time from the dying instance.
- **Fix:** once a relaunch is under way the shutting-down instance suppresses all further on-leave prompts (it has already saved), so no stray dialog is left behind the splash.

Long-standing glitch, unrelated to #80. Release build verified.

#82 Dev playbooks distilled to one-command scripts

File	+	−
`tools/add-changelog-entry.ps1`	171	0
`tools/push-all.ps1`	79	0
`tools/regen-overview-pdf.ps1`	54	0
`tools/gh.ps1`	38	0
`docs/playbooks/*.md (5 files)`	42	11

The mechanical playbook procedures — the ones that are a fixed command sequence rather than judgement — are now small scripts in `tools/`. The value is retrieval over re-derivation: one path plus args instead of reconstructing the command (and its failure traps) every session. The playbooks stay as the invocation note and keep the gotchas.

- **tools/push-all.ps1** — pushes the work all the way out: `origin/integration` *and* `v2/master`, with an ahead/behind report, a clean-tree guard, and a verify that both refs actually landed (`-DryRun` to preview).
- **tools/regen-overview-pdf.ps1** — regenerates `Overview.pdf` via headless Edge with a fresh `--user-data-dir` each run (without it Edge silently no-ops) and a LastWriteTime check that fails loudly if it did.
- **tools/gh.ps1** — runs the portable `gh.exe` with `GH_TOKEN` injected from the registry (harness shells do not inherit it), args passed straight through.
- **tools/add-changelog-entry.ps1** — adds a `#N` entry to this changelog from a JSON spec: it derives the number, computes every count from the file rows and re-sums the grand totals, inserts the detail block, TOC row and at-a-glance row, bumps the header count, and self-checks that all four places agree. You author only the content — this very entry was generated by it.

Tooling only; no application change, so nothing to build.

#83 Relaunch supervisor, true single-instance & crash reporting

File	+	−
ioSender XL/ioSender XL/CrashReporter.cs	584	0
CNC AppLaunch/CNC AppLaunch/AppLaunch.cs	106	33
CNC Controls/CNC Controls/PipeServer.cs	51	25
ioSender XL/ioSender XL/App.xaml.cs	48	0
CNC Core/CNC Core/RelaunchSupervisor.cs	38	0
ioSender XL/ioSender XL/MainWindow.xaml.cs	13	0
CNC Controls/CNC Controls/GrblConfigView.xaml.cs	10	0

Three linked robustness features around the app's launch, restart and crash lifecycle. `AppLaunch.exe` — historically the single-instance file-open forwarder — is repurposed as a small, application-agnostic **relaunch supervisor**: `AppLaunch.exe -ExitCodeToRelaunch 240 -- <exe> [args]` starts the child with an `EXITCODE_TO_RELAUNCH` environment variable set and relaunches it whenever it exits with exactly that code (any other code propagates and ends the loop). The env var is the whole handshake; ioSender knows nothing about AppLaunch beyond it.

- **True single instance.** On startup ioSender probes its named pipe: if another instance is already running it forwards the file argument (if any), raises that window, and exits — a second launch never opens a second window. This moved out of AppLaunch and into ioSender, where the server already lived.
- **Clean supervised restart.** When supervised, the in-app “restart to apply” simply exits with code 240 and lets the supervisor relaunch it — no in-process re-spawn, so the restart prompt can no longer strand itself behind the relaunch splash. Unsupervised runs (development, headless) keep the in-process relaunch as a fallback.
- **Opt-in crash reporting, no backend.** A fatal unhandled exception now writes a real minidump plus a text summary to `%AppData%\ioSender\crashes`. The next launch shows a consent dialog with the summary; “Send report” bundles the dump into a `.zip` (GitHub rejects raw `.dmp`), opens a pre-filled GitHub issue, and docks the file-manager window to a strip across the top of the screen with the zip selected, ready to drag onto the issue — sized in physical pixels and raised past the foreground lock so it works correctly on high-DPI displays. A topmost instruction banner spells out the one step to take.

Builds clean (Debug + Release). Wiring the desktop shortcut / file association to AppLaunch is a packaging step (no installer in this repo). Hidden `-crashtest` flag exercises the crash → report path.

#84 Main-menu overhaul: dissolve the File menu, slim the menu bar

File	+	−
ioSender XL/ioSender XL/MainWindow.xaml.cs	47	37
CNC Controls Camera/CNC Controls Camera/ConfigControl.xaml.cs	43	0
CNC Controls/CNC Controls/GCodeListControl.xaml.cs	29	0
CNC Controls/CNC Controls/AppConfig.cs	27	1
ioSender XL/ioSender XL/ProgramPanel.xaml	26	6
ioSender XL/ioSender XL/ProgramPanel.xaml.cs	19	0
CNC Controls/CNC Controls/Converters.cs	15	0
CNC Controls Camera/CNC Controls Camera/ConfigControl.xaml	14	0
ioSender XL/ioSender XL/MainWindow.xaml	13	20
ioSender XL/ioSender XL/JobWorkspace.xaml.cs	12	0
CNC Controls Camera/CNC Controls Camera/Camera.xaml.cs	11	0
CNC Controls/CNC Controls/GCode.cs	7	0
CNC Controls/CNC Controls/GCodeListControl.xaml	6	1
CNC Controls/CNC Controls/ICamera.cs	1	0
Locale/*/csv (7 locales x 3 assemblies)	188	0

The menu bar is slimmed to essentials and the File menu is dissolved, with each item rehomed to where it is actually used.

- **Program-view header** â€" with no file loaded it offers Load File / Load Folder; once loaded it shows the program source with a close (Ã—). Save and a Transform submenu move to the program list's right-click menu â€" Save works on any program view, Transform is rebuilt per menu-open.
- **File menu gone** â€" Connect/Reconnect is a top-level item, Open Console is a double-click on the Console tab (Esc still toggles it), Open Application data folder moves to a new Help â†' Support submenu, and Exit is dropped (the window close / Alt+F4 quit).
- **Toolbar row removed** â€" the file-action icons and quick-run macro buttons beneath the menu bar are gone; macros still run by F-key.
- **Camera is opt-in** â€" instead of always showing (and opening a laptop's built-in webcam), the Camera menu appears only when a video device is bound via a new Device picker + Connect/Disconnect button in Settings â†' App â†' Camera.

Builds clean Debug + Release; user-verified. Localized across all 7 locales. Commits 52b51d1 / 54ebcdd / 9511d3e.

#85 Start Job corner probe: one macro + sacrificial spacer/backer support

File	+	−
macros/pcorner.macro	21	14
macros/probe_tfl.macro	0	147
macros/cal.macro	0	3
macros/start_job.macro	8	49
ioSender XL/ioSender XL/StartJobView.xaml.cs	9	3
ioSender XL/ioSender XL/StartJobView.xaml	2	0
CNC Controls/CNC Controls/StartJobConfig.cs	5	0
CNC Controls/CNC Controls/AtcMacros.cs	6	6
CNC Controls/CNC Controls/CNC Controls.csproj	0	2
SDCardView / MachineSetupWizard / JobView / GrblViewModel (string + comment cleanup)	8	8
Locale/*/csv (7 locales)	21	0

Start Job's corner probe is now a single macro, and it understands a sacrificial spacer/backer board under the stock.

- **One probe macro.** Dropped `cal.macro` and `probe_tfl.macro`; the sole corner probe is `pcorner.macro` — a superset (all four corners, tip radius and feeds from the 3D probe definition, Z soft-limit clamp, per-corner flatness). The live Start Job program already emitted `pcorner`, so nothing changed at run time; the old pair survived only as a pre-Generate seed and is gone, leaving fewer files to install on the controller.
- **Sacrificial spacer/backer.** A new optional *Spacer/backer (Z)* field covers the thin-sheet case — e.g. a 2.5 mm aluminium sheet taped to a 1/4 in MDF backer of the same footprint. The probe finds the real spoilboard in the clear air beside the stock, so `pcorner` now derives an effective floor = spoilboard + spacer and starts the face probe from there: the 2 mm tip lands on the metal instead of dropping into the backer/tape band. A spacer of 0 reproduces the previous spoilboard-only behaviour exactly, so bare-fence jobs are unaffected.

Hardware-validated on a grblHAL Teensy 4.1 rig: the 2 mm probe tip lands on a 2.5 mm sheet edge sitting on a 1/4 in backer, and a spacer of 0 leaves normal-stock probing unchanged.

#86 Start Job: fail fast when a corner is over bare board

File	+	−
macros/pcorner.macro	12	4

With *Measure stock size* on, corners 2–4 are positioned from the estimated stock size. If that estimate is too large a corner lands over bare spoilboard — and previously the top probe there sought all the way down to the board, mistook it for the stock top, then drove a side probe into empty air that only faulted at the search limit. Worse, a Feed-Hold then Stop during that latch could leave the controller needing a reset.

Those reuse corners now cap the top probe at the effective floor (spoilboard + spacer): a genuine corner still triggers on the material above it, but a corner over bare board reaches the floor with nothing there and the probe aborts immediately — no side search into air, and nothing to wedge. The first corner, positioned by hand from G28, still discovers the board as before, so flatness/skew measurement is unaffected.

#87 In-app UI test server (flag-gated developer / QA harness)

File	+	−
`ioSender XL/UiTestServer.cs`	472	0
`ioSender XL/App.xaml.cs`	23	0
`ioSender XL/MainWindow.xaml.cs`	10	1

A small in-process HTTP server, **off by default** and started only with `-testserver[=PORT]` or `IOSENDER_TESTSERVER=1|PORT` (default port 8760), that lets an external script drive the running UI and read state back — so a change can be exercised in the real app instead of always being handed off for someone to click. A normal user run never opens a socket.

- **Addressing by x:Uid.** Localization already put a unique `x:Uid` on nearly every interactive control; the server surfaces those at runtime (`UIElement.Uid`), walks the visual tree of the open windows, and acts through each element's *AutomationPeer* — the same machinery real UI-automation uses, so it respects enabled/disabled state and fires the real handlers.
- **Routes (JSON):** `GET /ping`, `/idle`, `/tree`, `/state/{uid}`; `POST /invoke/{uid}`, `/set/{uid}?value=`.
- **Nav tabs are addressable too.** The top-level tabs are built in code, so they carried no `x:Uid`; they now get one from the tab registry, and selecting a tab realizes its (lazily-created) content.
- **No new dependencies, no admin.** A raw loopback `TcpListener` (not `HttpListener`, which would need a `netsh` URL reservation) keeps it zero-config for unattended runs.

Debug and Release verified; the server is inert unless explicitly enabled.

#88 Machine Setup no longer false-flags “install ATC macros” on connect

File	+	−
CNC Controls/AtcMacros.cs	9	1
CNC Controls/SDCardView.xaml.cs	8	3

On connecting to an ATC controller, the Machine Setup gate could jump straight to step 6 (“install ATC macros”) even though the macros were already present — the step then painted green a moment later. A timing race between the setup check and the first controller-filesystem listing.

- **Cause.** The macro-status check lists the controller filesystem, which pumps the dispatcher and has a re-entrancy guard. When another listing was already in flight (just after it cleared the shared file table), the status check's own listing was skipped and it read a half-cleared table, so the required macros looked absent — and an absent-looking result was reported as *Missing*, which the gate read as “not installed.”
- **Fix.** The filesystem loader now reports whether it actually performed a listing; when it did not (skipped, or the link was down) the status check returns *unknown* rather than fabricating *Missing* rows. The gate already treats unknown as “no action,” and a genuinely-empty filesystem still lists a placeholder, so a real first-run *missing* case still routes to step 6 correctly.

#89 UI test server matured into a full automation harness

File	+	−
`WpfUiTestServer/ (new assembly: engine, status seam, HTTP spec)`	1371	0
`CNC Core/AppDialogs.cs`	92	0
`ioSender XL/GrblStatusProvider.cs`	53	0
`ioSender XL/App.xaml.cs`	63	3
`MessageBox.Show → AppDialogs (7 projects, ~130 sites)`	135	130
`x:Uid on named controls (52 XAML files)`	112	105
`ioSender XL/MainWindow.xaml.cs`	6	3

The flag-gated UI test server introduced in #87 was built out into a capable automation harness and lifted into its own **app-agnostic assembly** (`WpfUiTestServer`, no ioSender references). It still only exists behind `-testserver` / `IOSENDER_TESTSERVER`; a normal run never opens a socket. Full HTTP reference in `WpfUiTestServer/README.md`.

- **Drive & assert.** Address any control by `x:Uid` (with `?index=` for duplicates); `invoke/set/select`; read `/state`, `/tree`, and a `/uids` discovery list (uid + type + count). App-level `/status` (connection, controller state, streaming/job, DRO) via a host-supplied provider seam, and `/waitfor` to block on a control or status condition rather than guessing at timing.
- **See it.** `GET /screenshot[/{uid}]` returns a PNG of the window or an element.
- **Dialogs no longer block automation.** Every message box now routes through `AppDialogs` (a `MessageBox.Show`-shaped wrapper — identical for a real user); the harness can pre-arm/answer prompts and read their text back. All ~130 `MessageBox.Show` sites across seven projects were migrated (the fatal crash box excepted).
- **Input.** Keyboard injection (`/key`) and context-menu open/invoke (`/menu`).
- **Failure visibility.** `GET /exceptions` surfaces unhandled/route exceptions the app used to swallow into a crash-and-exit, and a small `ioSender.exit.json` reports the exit code + reason after the app goes away.
- **Addressability sweep.** Added `x:Uid` to every named, addressable control that lacked one (105 across 52 files), plus the generated nav tabs and Machine-Setup step tabs, so the whole UI is reachable.
- **Operator safety.** A pulsing red banner across the top marks when the app is under test-server control.

Off by default; loopback only, no auth — keep it test/dev-gated. New x:Uids on a few text-bearing controls await a localization CSV pass (English fallback works). Debug + Release verified.

#90 WpfUiTestServer extracted to a standalone NuGet package

File	+	−
`WpfUiTestServer/* — extracted to standalone repo/package`	0	1371
`ioSender XL/ioSender XL/ioSender XL.csproj`	1	4
`CNC Core/CNC Core/CNC Core.csproj`	2	5
`ioSender XL/ioSender XL.sln`	0	10
`build.ps1`	2	1

The app-agnostic UI test server assembly introduced in #89 now lives in its own public repository (`github.com/stevenrwood/WpfUiTestServer`) and ships as a NuGet package. ioSender references it like any third-party dependency; the in-repo copy is gone. No user-facing behaviour change — it remains off by default, behind `-testserver` / `IOSENDER_TESTSERVER`.

- **Package, not project.** Both consumers (`ioSender XL` and `CNC Core`) swapped their `ProjectReference` for `<PackageReference Include="WpfUiTestServer" Version="1.0.0">`; the project was dropped from the solution and the folder removed.
- **Published via Trusted Publishing.** The standalone repo packs and publishes to NuGet from GitHub Actions using OIDC (no stored API key), converted to an SDK-style `net462` project with package metadata.
- **Build now restores.** These projects had no prior NuGet dependencies, so `build.ps1` gained `-restore` on the MSBuild call. Building in Visual Studio restores automatically; a raw `msbuild` of the solution now needs a restore step too.

Dev/QA infrastructure only. Debug + Release both build-verified against the published package.